# Forecast Uncertainty, Model Assumptions, and Fiscal Targets in the EU Debt Sustainability Analysis

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**Abstract**

The reformed EU fiscal rules derive each country's required fiscal adjustment from a debt sustainability analysis (DSA). Using a replication of the Commission's 2024 reference trajectories, this paper quantifies how much the resulting targets for the structural primary balance depend on the inputs of the DSA. Because the target is the highest balance required across several criteria, an input matters only where it affects the criterion that binds. I draw forecasts and model parameters jointly from past forecast errors and published estimates. Across the 15 countries that receive a reference trajectory and have adjustment needs, the interquartile range of the binding target averages 1.1 percentage points of GDP, running from 0.39 points below the baseline to 0.68 points above it, against an average required adjustment of 3.3 points over four years. With multipliers above one, the debt sustainability safeguard requires implausibly large adjustments over four years. The stochastic criterion rarely binds, so its specification, with the exception of winsorisation, matters little. How long the rules require governments to cover rising ageing costs shifts the DSA targets by about as much as forecast errors, whereas the probability threshold and the size of the stress tests shift them less.

**JEL classification:** E62, H62, H63, H68

**Keywords:** debt sustainability analysis, EU fiscal rules, sensitivity analysis, forecast errors, fiscal multiplier

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# 1 Introduction

Since April 2024, the EU fiscal rules have set fiscal targets for each member state through a debt sustainability analysis (DSA). Member states with government debt above 60 percent of GDP or a deficit above 3 percent of GDP receive a reference trajectory from the European Commission, which specifies a path for net expenditure over an adjustment period of four to seven years. The Commission sent these trajectories to governments in June 2024 as prior guidance for their medium-term fiscal-structural plans; all other member states receive non-binding technical information showing the adjustment implied by the DSA. Each trajectory translates into a target for the structural primary balance (SPB), defined as the primary balance corrected for the economic cycle and one-off measures, at the end of the adjustment period. The Commission sets the target so that over the ten years after the adjustment, the debt ratio declines or stays below 60 percent of GDP in a baseline projection, in three stress tests and in a five-year stochastic simulation, while the headline deficit stays below 3 percent of GDP. Numerical safeguards and the excessive deficit procedure (EDP) can raise the target further.

While the DSA tailors these targets to country-specific conditions, it also makes them depend on many forecasts, parameters and conventions. Governments can argue for different inputs in their medium-term plans, at times with large consequences for the required adjustment (Zettelmeyer et al., 2025). The methodology itself is under review. In the Debt Sustainability Monitor 2025, the Commission and the member states agreed its first changes: a lower fiscal multiplier and a new method for the stochastic projections (European Commission, 2026). Both assessing national plans and reviewing the framework require knowing how much the targets depend on each input.

This paper quantifies that dependence for all 27 member states. It uses the replication of the Commission's 2024 prior guidance from Darvas et al. (2023, 2024), which can be checked directly against the targets published by the Commission. I divide the inputs into four groups: forecasts, model parameters, estimation choices in the stochastic analysis and calibration conventions. I vary the forecasts and the parameters one at a time and jointly, drawing them from historical forecast error distributions and published estimates. Because the target is the highest SPB required by several criteria, an input alters it only through the criterion that binds, so favourable and adverse inputs have unequal effects (Section 2.3). Headline results are reported for the 15 countries that receive a reference trajectory and whose baseline target requires a positive adjustment.

Four results stand out. First, when forecasts and parameters are drawn jointly from historical ranges, the interquartile range of the binding target averages 1.07 percentage points of GDP. This is the range of targets that a government could defend with inputs inside past error ranges, and it compares with an average required adjustment of 3.3 points over four years. Second, the fiscal multiplier matters mainly through the debt sustainability safeguard, which requires debt to fall during the adjustment period itself. With multipliers above one, this safeguard requires

implausibly large adjustments over four years or cannot be met. Third, the stochastic criterion rarely binds, because the adverse interest-growth ($r - g$) stress test requires a higher SPB in 12 of the 15 countries. The specification of the stochastic analysis therefore matters little, with the exception of the winsorisation of shocks. Removing winsorisation raises the DSA targets of Austria and France by 0.66 and 0.57 points respectively. Fourth, dropping the ten-year window over which the target covers rising ageing costs lowers the average DSA target by 0.59 points, more than raising the probability threshold and all stress-test shocks together (+0.25 points).

These findings suggest three priorities for the review of the framework. First, the ageing-cost window and the deterministic stress tests deserve at least as much attention as the stochastic simulations. Second, the required decline in debt should be measured over a window after the adjustment, or the debt sustainability safeguard dropped, because testing debt reduction during the adjustment years makes the targets depend heavily on the multiplier and on any temporary output loss. Third, the degree of risk that the rules accept through stress tests, probability thresholds and long-run conventions should be justified explicitly.

The paper adds to a growing literature evaluating the reformed framework and the role of the DSA within it (Heimberger, 2023; Pench, 2024, 2025; Boivin and Darvas, 2025; European Fiscal Board, 2025). Four studies for the European Parliament assess the DSA methodology. Cottarelli (2024) reviews the stochastic method and shows that it ties the required annual decline in debt to the estimated volatility of debt, from 0.80 percent of GDP in France to 2.37 percent in Greece. Wieland (2024) enters the assumptions of Italy's national plan into the Commission's calculation sheet, lowering projected debt in 2030 by 5 percentage points of GDP, and argues that the stress tests miss risks during the adjustment period. Erce (2025) and Jousten (2025) discuss the key assumptions qualitatively, with Erce (2025) showing that revisions to France's estimated SPB between the 2022 and 2024 forecasts average 0.7 percent of GDP. Gros and Hofer (2026) derive simplified rules of thumb for the DSA criteria and check them against simulations of our replication of the Commission's DSA. Related work examines single inputs in isolation, including the fiscal multiplier (Heimberger et al., 2024), demographic projections (Darvas et al., 2025b), growth assumptions in Germany's plan (Zettelmeyer et al., 2025), the stochastic DSA (Darvas et al., 2025a; Bec et al., 2025) and the effect of public investment on potential growth (Gottschalk et al., 2025). This paper quantifies the effects of all forecasts, parameters, estimation choices and calibration conventions in a single framework, calibrating the input variations to past forecast errors and published estimates.

Section 2 describes how the DSA sets the targets and classifies the inputs. Section 3 presents the historical forecast error distributions and reports the effects of the forecasts and model parameters. Section 4 examines the fiscal multiplier and the debt sustainability safeguard. Sections 5 and 6 turn to the estimation choices and the calibration conventions. Section 7 reports robustness checks, and Section 8 concludes.

# 2 Fiscal adjustment targets

## 2.1 Debt sustainability analysis in the EU fiscal rules

The baseline of this analysis is the replication of the Commission's DSA presented in Darvas et al. (2024). The model projects the debt-to-GDP ratio with the standard debt-dynamics identity (European Commission, 2024b), where nominal growth, the primary balance and the average interest rate on debt depend on the SPB path through a fiscal multiplier and the budget semi-elasticity (Carnot and de Castro, 2015; Mourre et al., 2019).[1] The Annex describes the model and the data in detail. The base year of the projection, the last year before the adjustment period, is defined as $T$. The baseline uses the data of the Commission's 2024 prior guidance calculation sheets, so $T$ is 2024 and a four-year adjustment runs from 2025 to 2028. Because the Commission published the targets implied by these sheets, the replication can be checked directly against them. The baseline also follows the Commission's 2024 method, with a fiscal multiplier of 0.75 and normally distributed shocks in the stochastic projections. Robustness tests include agreed updates to the methodology and 2026 data vintages.

The DSA sets the target as the lowest SPB at the end of the adjustment period that meets a set of solvency and deficit criteria. I distinguish two targets. The *DSA target* is the lowest SPB that meets the criteria of the DSA alone. The *binding target* is the DSA target, raised where needed to meet the safeguards, the EDP and the adjustment floor described below. The DSA comprises three types of criteria. Under the deterministic debt criteria, the debt ratio must decline in each of the ten years after the adjustment period, or stay below 60 percent of GDP at the end of those ten years. This must hold in a baseline projection and in three stress tests, which assume a lower SPB, a higher interest-growth differential ($r - g$) and a temporary rise in market interest rates. Under the deficit criterion, the headline deficit must stay below 3 percent of GDP from the last adjustment year until ten years after the adjustment period. Under the stochastic criterion, debt five years after the adjustment period must be at or below its level at the end of the adjustment period with a probability of at least 70 percent, computed from simulated shocks to growth, interest rates, exchange rates and the primary balance.

To move from the DSA target to the binding target, four additional constraints apply. First, an adjustment floor prevents favourable trajectories from implying fiscal expansion. Second, under the EDP, the annual adjustment must be at least 0.5 percentage points of GDP after each year with a deficit above 3 percent of GDP, measured in the SPB until 2027 and in the structural balance from 2028. Third, the debt sustainability safeguard requires the debt ratio to decline during adjustment years outside the EDP, by an average of at least 1 percentage point per year while debt exceeds 90 percent of GDP, and at least 0.5 points while it lies between 60 and 90 percent. Fourth, the deficit resilience safeguard requires the SPB to improve by at least 0.4 percentage points per year (0.25 points with a seven-year adjustment) as long as the structural

[1] The budget semi-elasticity is the change in the budget balance, in percent of GDP, when output changes by 1 percent.

**Table 1:** Baseline SPB targets and Commission prior guidance, four-year adjustment (percent of GDP)

| | 2024 | | DSA target | | Binding target | | | |
|---|---|---|---|---|---|---|---|---|
| | Debt | SPB | Comm. | Model | Comm. | Model | DSA criterion | Binding constraint |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Greece | 153.9 | 1.73 | 2.37 | 2.30 | 2.61 | 2.55 | Adverse $r-g$ test: debt declines | Deficit resilience |
| Italy† | 138.6 | −1.05 | 3.35 | 3.30 | 3.35 | 3.30 | Adverse $r-g$ test: debt declines | DSA criteria |
| France† | 112.4 | −2.97 | 0.79 | 0.87 | 0.79 | 0.87 | Deficit below 3% | DSA criteria |
| Spain† | 105.5 | −0.85 | 2.59 | 2.56 | 2.59 | 2.56 | Adverse $r-g$ test: debt declines | DSA criteria |
| Belgium† | 103.4 | −2.03 | 1.25 | 1.18 | 1.25 | 1.18 | Adverse $r-g$ test: debt declines | DSA criteria |
| Portugal | 95.6 | 2.20 | 2.52 | 2.49 | 2.52 | 2.49 | Adverse $r-g$ test: debt declines | DSA criteria |
| Finland† | 80.5 | −0.53 | 2.11 | 1.81 | 5.87 | 5.97 | Stochastic | Debt safeguard |
| Austria† | 79.5 | −1.34 | 0.90 | 0.81 | 2.58 | 2.58 | Adverse $r-g$ test: debt declines | Debt safeguard |
| Hungary | 74.3 | 0.03 | 2.67 | 2.65 | 2.67 | 2.65 | Adverse $r-g$ test: debt declines | DSA criteria |
| Cyprus | 70.6 | 3.48 | 3.48 | 0.33 | 3.48 | 3.48 | Deficit below 3% | Adjustment floor |
| Slovenia | 68.1 | −1.24 | 0.52 | 0.47 | 0.52 | 0.47 | Lower-SPB test: debt below 60% | DSA criteria |
| Germany† | 62.5 | −1.02 | 1.14 | 1.07 | 2.74 | 2.77 | Lower-SPB test: debt below 60% | Debt safeguard |
| Croatia | 59.5 | −1.96 | – | −0.50 | −0.36 | −0.36 | Baseline: debt below 60% | Deficit resilience |
| Slovakia | 58.5 | −4.26 | 0.94 | 0.89 | 0.94 | 0.89 | Lower-SPB test: debt below 60% | DSA criteria |
| Poland | 53.7 | −2.61 | 0.67 | 0.74 | 0.67 | 0.74 | Deficit below 3% | DSA criteria |
| Malta | 52.0 | −2.90 | −1.14 | −1.09 | −0.87 | −0.87 | Deficit below 3% | EDP minimum adjustment |
| Romania† | 50.9 | −4.41 | 0.47 | 0.54 | 0.47 | 0.54 | Deficit below 3% | DSA criteria |
| Netherlands | 47.1 | −0.64 | – | 0.07 | 0.08 | 0.07 | Deficit below 3% | DSA criteria |
| Czechia | 45.2 | −0.09 | – | 0.38 | 0.39 | 0.38 | Deficit below 3% | DSA criteria |
| Latvia | 44.5 | −1.36 | – | −1.27 | −0.14 | −0.14 | Deficit below 3% | Deficit resilience |
| Ireland | 42.5 | 2.53 | – | −0.66 | – | −0.66 | Deficit below 3% | DSA criteria |
| Lithuania | 38.3 | −0.37 | – | 0.25 | 0.27 | 0.25 | Deficit below 3% | DSA criteria |
| Sweden | 32.0 | 0.69 | – | −1.77 | −0.83 | −0.75 | Deficit below 3% | Deficit resilience |
| Luxembourg | 27.1 | 0.10 | – | −0.23 | – | −0.23 | Deficit below 3% | DSA criteria |
| Denmark | 26.5 | 2.91 | – | −1.41 | −1.09 | −1.09 | Deficit below 3% | Adjustment floor |
| Bulgaria | 24.5 | −2.09 | – | −2.07 | −0.87 | −0.89 | Deficit below 3% | Deficit resilience |
| Estonia | 21.4 | −0.29 | −0.29 | −1.84 | −0.29 | −0.29 | Deficit below 3% | Adjustment floor |

Source: Author's calculations based on the European Commission's prior guidance calculation sheets of 2024. Note: SPB at the end of the adjustment period (2028). Columns (3) and (5) are the Commission's targets, and columns (4) and (6) the model baseline. Column (3) is the SPB in 2024 plus four times the DSA-based annual adjustment; countries receiving technical information report no DSA-based adjustment. For Cyprus and Estonia, the Commission reports the adjustment floor (the 2024 SPB) in column (3). For Denmark, which receives technical information, the floor lies 1 percentage point per year below the 2024 SPB and sets the binding target in columns (5) and (6). Throughout the paper, the model's DSA target in column (4) and its setting criterion in column (7) exclude the adjustment floor. The Commission's targets round the annual adjustment up to 0.01 percentage points, whereas the model keeps exact values. Column (8) is the constraint that sets the binding target: the DSA criteria, the adjustment floor, the debt sustainability safeguard (debt safeguard), the deficit resilience safeguard or the minimum adjustment of the EDP. Countries are ordered by debt in 2024. Table A2 in the Annex reports the seven-year targets. †Medium-term plan with a seven-year adjustment period.

deficit exceeds 1.5 percent of GDP (Annex A.3).

I implement these rules as in the Commission's prior guidance calculation sheets, which have several notable features. First, the deterministic debt criteria apply in one of two variants to all scenarios at once: either debt declines in the baseline and the three stress tests and with at least 70 percent probability in the stochastic analysis, or debt stays below 60 percent of GDP ten years after the adjustment in the baseline and the three stress tests. The less demanding variant applies. Second, the annual adjustment is constant across the adjustment period, although the rules also permit a front-loaded path. Third, the adjustment floor prohibits negative required adjustments for countries receiving a reference trajectory and sets a floor of −1 percentage point per year for countries receiving technical information. Fourth, countries with debt below 60

percent of GDP and a deficit below 3 percent receive technical information, which reports the SPB needed to keep their deficit below 3 percent and their debt below 60 percent ten years after the adjustment in the baseline scenario alone.

Unless stated otherwise, results in this paper use the 2024 data and method, a four-year adjustment and the 15 reference-trajectory countries with positive adjustment needs (excluding the ten technical-information countries and Cyprus and Estonia, where the adjustment floor binds).

## 2.2 Replication of the Commission targets

Table 1 compares the baseline model output with the Commission's published targets. Across member states, the mean absolute difference from the Commission's DSA targets (including the adjustment floor as the Commission reports them) is 0.07 percentage points and the median is 0.05 points. The largest difference, 0.30 points for Finland, arises in the stochastic criterion and stems from the width of the simulated debt distribution: five years after the adjustment, the 70th percentile of Finland's debt lies 3.9 points above the median in the model, compared with 5.2 points in the Commission's calculation sheet. The second largest difference, 0.09 points for Austria, arises in the adverse $r - g$ stress test. For the binding target, the mean absolute difference is 0.03 points, the median is 0.02 points and the largest is 0.10 points, for Finland.[2] In the three countries where the debt sustainability safeguard sets the binding target (Finland, Austria and Germany), the model yields 5.97, 2.58 and 2.77 percent of GDP, against 5.87, 2.58 and 2.74 percent in the Commission's sheets. For a seven-year adjustment, the mean absolute difference is 0.09 points for the DSA target and 0.08 points for the binding target (Table A2 in the Annex).

## 2.3 Determining the binding criterion

Each criterion requires its own minimum SPB, and the DSA target is the highest of these requirements. Figure 1 shows the SPB required by each DSA criterion and each constraint of the binding target in the baseline. Across the 27 member states, the deficit criterion sets the DSA target for 15 countries, the adverse $r - g$ stress test for seven, the lower-SPB stress test for three, the baseline debt criterion for one and the stochastic criterion for Finland. For Cyprus, Estonia, Slovenia, Germany, Slovakia, Malta and Romania, the variant keeping debt below 60 percent of GDP is less demanding and therefore applies. In 12 member states, a constraint outside the DSA criteria sets the binding target: the debt sustainability safeguard in Finland, Austria and Germany; the deficit resilience safeguard in Greece, Croatia, Latvia, Sweden and Bulgaria; the minimum adjustment of the EDP in Malta; and the adjustment floor in Cyprus, Denmark and Estonia.

[2] Unlike the Commission's sheets, the model does not round the annual adjustment up to 0.01 percentage points, which accounts for differences of up to 0.04 points over four years (Annex A.3).

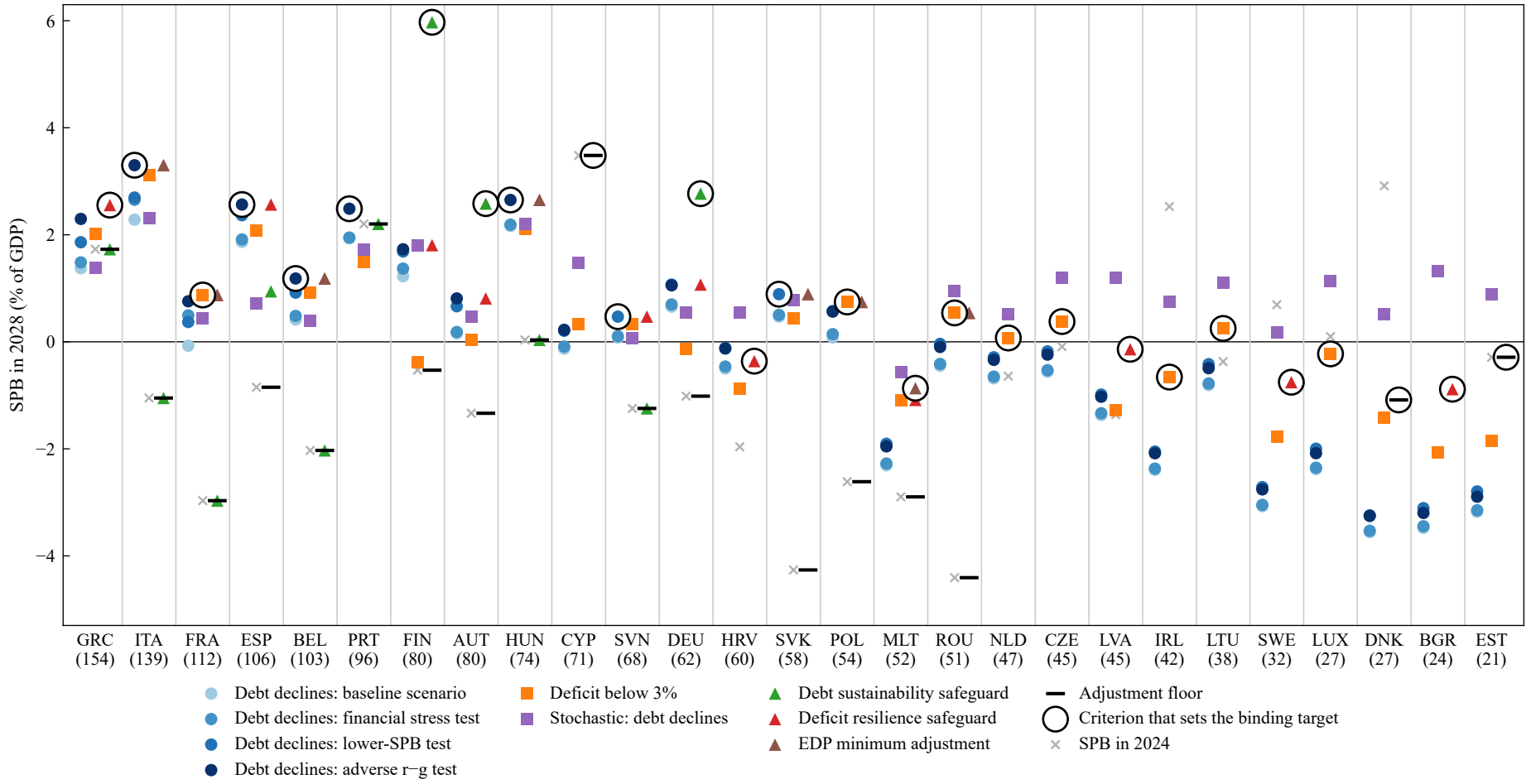



**Figure 1:** SPB required by each criterion in the baseline, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: SPB at the end of the adjustment period (2028). Each column shows one country, ordered by debt in 2024 (in brackets), with the debt criteria (circles), the deficit and stochastic criteria (squares) and the constraints of the binding target (triangles) side by side. The open circle marks the constraint that sets the binding target. The bar shows the adjustment floor for reference-trajectory countries or where it binds. Grey crosses show the SPB in 2024. The debt criteria are shown in the variant that applies to the country. For the debt sustainability safeguard, the figure plots the standalone SPB required for reference-trajectory countries with debt above 60 percent of GDP. For the deficit resilience safeguard and the EDP, it adds their minimum steps to the DSA path. The adjustment floor equals the 2024 SPB for reference-trajectory countries and lies 4 percentage points below the 2024 SPB for technical-information countries.

Because the target is the maximum across criteria, an input changes the DSA target only if it moves the criterion that currently binds or lifts another criterion above it. Inputs that act only on slack criteria leave the target unchanged. This maximum operator makes the targets respond asymmetrically. As Figure 1 illustrates, when competing criteria sit close together, for example the deficit criterion and the adverse $r - g$ test in France and Belgium, an adverse input raises the target one-for-one, whereas a favourable input lowers the target only until the next criterion takes over as a floor. To evaluate each input both in isolation and within the full set of rules, I report its effect at three levels: the SPB required by the input's *standalone criterion* (the stress test itself for stress-test shocks, the stochastic criterion for stochastic settings and the baseline debt criterion for forecasts and model parameters), the *DSA target* and the *binding target*.

## 2.4 Classification of the inputs

Table 2 lists the inputs of the DSA in four groups, together with where each input enters the projection, its baseline value and the alternative values checked. The groups separate empirical uncertainty about the central projection (groups 1 and 2) from choices in estimating the risk around it (group 3) and conventions on how much risk the rules accept (group 4).

The central projection is the baseline deterministic scenario. It rests on medium-term forecasts of market interest rates, inflation, potential growth and ageing costs, and on model parameters such as the fiscal multiplier and the budget semi-elasticity. If these forecasts or parameters are off, the entire baseline debt path shifts, affecting every deterministic criterion and safeguard at once. I vary the forecasts in line with historical forecast error distributions and the parameters within the range of published estimates.

The third group contains the estimation choices of the stochastic analysis. In the Commission's 2024 method, shocks are drawn from a joint normal distribution fitted to historical changes in growth, interest rates, exchange rates and the primary balance, and added to the central projection to construct a five-year fan chart of debt paths. The 2026 method instead resamples the residuals of autoregressive models fitted to the same changes (see Annex A.2 for details). In my baseline implementation of both methods, the shocks have a mean of zero, so the fan chart stays centred on the central projection and captures year-to-year volatility rather than persistent baseline errors. Varying the shock model, the start of the shock sample, the shock frequency and winsorisation changes the width and shape of the fan chart while leaving all deterministic criteria unchanged.

The fourth group contains the calibration conventions. The probability threshold of the stochastic criterion and the shock sizes of the stress tests set margins against unexpected shocks to the baseline projection, while the start year and horizon of the stochastic projection determine how much of the fan chart the threshold evaluates. The ageing-cost window determines how many years of projected changes in ageing costs the SPB target must pre-fund, and the convergence of total factor productivity (TFP) growth after $T + 10$ assumed in the Ageing Report

credits countries that have weak productivity growth with an automatic long-run recovery. Both the ageing projections and the TFP convergence rule are agreed in working groups of the EU's Economic Policy Committee. The table also includes a data check replacing the Commission's refinancing assumptions with Eurostat data on debt by residual maturity (Section 7).

Three features of the baseline are held fixed across all checks. The first is the starting point of the projection (the SPB, the output gap and debt in 2024), which comes from the Commission's spring 2024 forecast for the current year and is subject to later revisions. The second is the stock-flow adjustment, which covers country-specific flows that change debt without passing through the deficit. In Finland, these flows mainly reflect the surpluses of public pension funds, which improve the general government balance but must be invested in financial assets and therefore do not reduce gross debt (Boivin and Darvas, 2025; Darvas et al., 2024); in Greece, they reflect the statistical recording of deferred interest on official loans.[3] Third, while this paper varies the inputs, parameters and conventions within the Commission's accounting framework, it holds the structure of the underlying model fixed, abstracting from extensions such as feedback from public investment to potential output (Gottschalk et al., 2025) or permanent output hysteresis.

[3] Because these flows follow from legal and statistical rules, I retain the Commission's projections. They explain why Finland's binding target is so high: setting Finland's post-2024 stock-flow adjustments to zero lowers its DSA target from 1.81 to 0.32 percent of GDP and its binding target from 5.97 to 0.99 percent.

**Table 2:** Inputs of the DSA by group

| Input | Where it enters | Baseline | Values checked |
|---|---|---|---|
| *1. Forecasts: distribution of past forecast errors* | | | |
| Real short-term rate, $T + 10$ | Interest on new and refinanced debt | Commission | ±1 pp. |
| Real long-term rate, $T + 10$ | Interest on new and refinanced debt | Commission | ±1 pp. |
| Inflation, $T + 10$ | Nominal growth and rates | Commission | ±0.5 pp. |
| Potential growth, $T + 10$ | Real growth | Commission | ±0.75 pp. |
| Real growth, $T + 1$ | Output in first adjustment year | Commission | ±1.25 pp. |
| Ageing costs, change 2024–2038 | SPB after the adjustment | Ageing Report | ±0.5 pp. of GDP |
| Ageing costs, projection | SPB after the adjustment | 2024 Ageing Report | 2021 Ageing Report |
| Ageing costs, demography | SPB after the adjustment | 2024 Ageing Report | Higher migration, lower fertility[e] |
| Rates and inflation, $T + 30$ | Long-run rates and growth | Commission | ±0.5 and ±0.25 pp.[a] |
| *2. Model parameters: range of published estimates* | | | |
| Fiscal multiplier | Output effect of the adjustment | 0.75 | 0.5 to 1.0; 0.6[b]; 1.5[c] |
| Persistence of the multiplier | Output effect of the adjustment | 3 years | 2 to 5 years |
| Output gap rule | Output effect of the adjustment | Recursive | Impulse-response[f] |
| Budget semi-elasticity | Cyclical balance | Commission | ×0.9 and ×1.1 |
| *3. Estimation choices: alternative choices* | | | |
| Shock model | Stochastic criterion | Normal | AR bootstrap, VAR bootstrap[d] |
| Start of shock sample | Stochastic criterion | 2000 | 1990, 2010 |
| Shock frequency | Stochastic criterion | Quarterly | Annual |
| Winsorisation of shocks | Stochastic criterion | 5th/95th pct. | 1st/99th pct., none |
| *4. Calibration conventions: alternative settings* | | | |
| Probability threshold | Stochastic criterion | 70% | 60%, 80% |
| Start of stochastic projection | Stochastic criterion | After adjustment | First adjustment year |
| Horizon of stochastic projection | Stochastic criterion | 5 years | 10 years |
| Adverse $r - g$ shock | Stress test | 0.5 pp. | 0.25, 0.75 pp. |
| Financial stress shock | Stress test | 1 pp. | 0.5, 1.5 pp. |
| Lower-SPB shock | Stress test | 0.5 pp. | 0.25, 0.75 pp. |
| Ageing cost period | SPB after the adjustment | 10 years | 5, 0 years |
| TFP growth after $T + 10$ | Potential growth | Ageing Report | Held at $T + 10$ value |
| *Data* | | | |
| Repayment profile of debt | Refinancing | Historical average | Eurostat residual maturity |

Note: Individual forecast shifts are deviations from the baseline, in percentage points (pp.), of about half the interquartile range of past errors (Table 3); the joint draws use the full error distributions. Forecasts and model parameters are varied one at a time and drawn jointly, except real growth in $T + 1$ (Section 4.3), the projections of the 2021 Ageing Report, the output gap rule and the values marked *a* to *c* and *e*, which are individual checks only. [a] Interest rates ±0.5 pp., inflation ±0.25 pp. [b] Multiplier of the Commission's 2026 method. [c] Crisis conditions. [d] Both also evaluated with the constants of the autoregressive models in the simulation, and the AR bootstrap (the Commission's method since the Debt Sustainability Monitor 2025) also without winsorisation. [e] Scenarios of the 2024 Ageing Report with 33 percent higher migration and 20 percent lower fertility. [f] Each step's output effect fades linearly over the persistence horizon of two to five years (Annex A.1).

**Table 3:** Distribution of past forecast errors

| Assumption | Source of errors | Sample | $N$ | Mean error | P10 | P25 | P75 | P90 |
|---|---|---|---|---|---|---|---|---|
| Real short-term rate, $T+10$ | Bundesbank: 1-year rate in ten years minus forward rate | Germany, year ends 1986-2015 | 30 | −4.13 | −1.61 | −1.22 | +0.40 | +3.11 |
| Real long-term rate, $T+10$ | Bundesbank: 10-year rate in ten years minus 10Y10Y forward rate | Germany, year ends 1986-2015 | 30 | −2.98 | −1.53 | −1.24 | +0.89 | +2.36 |
| Inflation, $T+10$ | IMF WEO: CPI inflation, average error over five years | 19 EU countries, autumn vintages 2000-2018 | 359 | +0.02 | −1.01 | −0.62 | +0.53 | +1.08 |
| Potential growth, $T+10$ | IMF WEO: potential growth forecast error (ex-post potential growth minus WEO forecast five years ahead) | 19 EU countries, autumn vintages 2000-2018 | 359 | −0.77 | −1.73 | −0.83 | +0.75 | +1.72 |
| Real growth, $T+1$ | IMF WEO: real GDP growth, error one year ahead | EU27, autumn vintages 2000-2018 | 513 | −0.12 | −2.81 | −1.07 | +1.47 | +2.76 |
| Change in ageing costs over 14 years | Ageing Reports 2015-2024: revision of the projected change in ageing costs over 14 years | EU27, 3 pairs of vintages | 81 | +0.01 | −1.05 | −0.59 | +0.44 | +1.00 |

Source: Author's calculations based on the autumn IMF World Economic Outlook vintages of 2000 to 2018 (International Monetary Fund, various years), as compiled by Ech-charfi (2024), the IMF World Economic Outlook database of October 2026 (International Monetary Fund, 2026), the Deutsche Bundesbank term structure of interest rates and European Commission (2015, 2018, 2021, 2024a). Note: Errors are realised minus forecast values, in percentage points. Potential growth forecast error: ex-post potential growth in the fifth year after each autumn WEO vintage (2000–2018), computed from real GDP and the output gap of the October 2026 WEO, minus the WEO forecast of real growth for the same year made in that vintage, demeaned by country across the 19 EU countries for which the WEO reports output gaps. Inflation: five-year average error of consumer price inflation for the same countries and vintages, demeaned by country. Real growth, $T+1$: error of the WEO forecast of real GDP growth for the year after each autumn vintage, demeaned by country across all 27 member states; it enters only the individual check of Section 4.3, not the joint draws. Forward-rate errors compare the German Bund forward rate at each year end (1986–2015) with the rate realised ten years later, demeaned. Ageing costs: revision to the change in total ageing costs over the 14 years after the base year of the newer report across consecutive report pairs (2015–2018, 2018–2021 and 2021–2024) and all 27 member states, in percentage points of GDP, not demeaned. Mean error is the average before demeaning; P10 to P90 are the percentiles of the demeaned errors (and of the revisions for ageing costs).

# 3 Forecasts and model parameters

## 3.1 Empirical distributions of forecast errors and parameter ranges

A forecast error is defined as the realised value minus the forecast. Table 3 reports the historical error distributions constructed for each variable (Annex A.5 provides further details on the underlying data). In the individual checks, I shift each forecast symmetrically by about half the interquartile range of past errors: ±1 percentage point for real interest rates, ±0.75 points for potential growth, ±0.5 points for inflation and ±0.5 points of GDP for ageing costs. Symmetric shifts show directly whether the targets respond more strongly to adverse errors than to favourable ones, while the joint draws of Section 3.4 sample from the full error distributions.

Because the DSA anchors medium-term growth at $T+10$, the relevant error is an error in medium-term potential growth. I measure it as the ex-post potential growth in a given year minus the forecast of real GDP growth for that year made five years earlier in the IMF's World Economic Outlook (WEO). By the fifth forecast year, the end of the WEO horizon, the output gap is projected to be closed, so the five-year-ahead real growth forecast is in effect a forecast of medium-term potential growth. Ex-post potential growth absorbs the permanent part of past

growth surprises because forecasting institutions revise their estimates of potential output after persistent shocks (Szörfi, 2026).

I remove each country's average error before sampling so that the ranges measure the dispersion of past errors rather than their historical bias. Before demeaning, the potential growth forecast error averaged −0.77 percentage points, reflecting the slowdown in trend growth over the sample period and perhaps forecast optimism of the kind Frankel and Schreger (2013) document for official euro-area forecasts. After demeaning, the interquartile range runs from −0.83 to +0.75 percentage points.[4]

For inflation, unlike growth, I use the average error over the five forecast years rather than the fifth-year error. Using individual year errors would inflate errors due to events like the energy price shock of 2022 and 2023. Calculated from WEO consumer price inflation forecasts for the same countries and vintages and demeaned by country, the interquartile range runs from −0.62 to +0.53 percentage points. These errors stand in for errors in the market-based inflation expectations that anchor the projection at $T + 10$.

For interest rates, the DSA uses ten-year market forward rates. To approximate errors in these forward rates, I compare ten-year-ahead forward rates on German government bonds with the spot rates realised ten years later. Forward rates exceeded realised rates by 3.0 percentage points on average, reflecting the term premium and the secular decline in nominal rates since the 1990s. Because raw historical errors combine overestimated interest rates (+3.0 points) with overestimated growth (+0.77 points), sampling un-demeaned errors from an era of falling rates would artificially flatter $r - g$ going forward. After removing the historical mean, the interquartile range runs from −1.24 to +0.89 percentage points for the long-term rate and from −1.22 to +0.40 points for the short-term rate. Because the German yield curve omits sovereign spreads over Germany, these ranges represent a lower bound on interest-rate risk for high-debt countries such as Italy and Greece, where detailed historical data were unavailable.

For ageing costs, I compare the baseline projections of the 2015, 2018, 2021 and 2024 Ageing Reports (European Commission, 2015, 2018, 2021, 2024a). For each pair of consecutive reports, the error is the revision to the projected change in total ageing costs over the 14 years after the newer report's base year, matching the 2024–2038 span of the four-year DSA. Across the 81 observations, the revisions average close to zero and have an interquartile range from −0.59 to +0.44 percentage points of GDP. While inter-report revisions understate uncertainty when successive reports err in the same direction and include pension reforms adopted between cycles, they provide an empirical benchmark for how much ageing projections move between updates. Because past revisions are unrelated to the initial level or projected change of ageing costs, I apply the same 0.5-point shift to all countries. Two further checks capture country-specific demographic risk by replacing each country's ageing-cost projection with its path under the 2024 Ageing Report scenarios assuming 33 percent higher migration or 20 percent lower fertility, both

[4]Lowering potential growth at $T + 10$ by the raw mean error of 0.77 percentage points raises the average DSA target by 0.40 percentage points of GDP, and Italy's by 0.94 points.

of which point in the direction of the latest Eurostat population projections, EUROPOP2025 (Haroutunian et al., 2026).

The ranges of the model parameters come from the empirical literature. The baseline multiplier is 0.75, as in Carnot and de Castro (2015) and the Commission's 2024 method. In 2026, the Commission lowered its multiplier to 0.6, citing an average of 0.62 across published estimates outside crises and periods at the zero lower bound, and of 0.50 in normal times alone (European Commission, 2026, Box 2.5). I vary the multiplier between 0.5 and 1.0 in the individual checks and joint draws, check the Commission's 2026 value of 0.6 separately, and test a value of 1.5 for crisis conditions outside the joint draws. Estimates for fiscal consolidations outside crises fall inside the 0.5 to 1.0 range, with differences largely driven by the composition of the consolidation,[5] whereas multipliers around 1.5 were estimated for the post-2010 European consolidations at the lower bound (Blanchard and Leigh, 2013) and in regional estimates within monetary unions (Nakamura and Steinsson, 2014; Chodorow-Reich, 2019). Because cumulative multipliers over two years are at least as large as impact multipliers (Barro and Redlick, 2011; Ramey and Zubairy, 2018) and consolidations can have lasting effects on output (Fatás and Summers, 2018; Gechert et al., 2019), I vary the persistence of the multiplier effect from two to five years around the three-year baseline. Alongside the persistence horizon, I also replace the recursive output gap rule of the Commission's sheets, under which two-thirds of the previous year's output gap deviation carries over during the adjustment, with a linear impulse-response rule in which the effect of each adjustment step fades independently over two to five years (Section 4.1 and Annex A.1). Finally, I scale the budget semi-elasticity by 0.9 and 1.1, matching the largest country revisions of about 10 percent between the Commission's 2014 and 2018 updates (Mourre et al., 2019).

## 3.2 Implementing shifts in the DSA

I shift real interest rates, inflation and potential growth at the $T + 10$ anchor. Each shift rises linearly from zero in the last short-term forecast year ($T + 1$) to its full size at $T + 10$, and then returns linearly to zero by $T + 30$, in line with the Commission's interpolation between medium- and long-run anchors.[6] Real GDP growth moves one-for-one with potential growth, leaving the baseline output gap unchanged, and nominal interest rates move one-for-one with inflation because the rate shifts are specified in real terms. The shift in ageing costs grows linearly from 2024 to reach its full size in 2038.

[5] In Guajardo et al. (2014), a consolidation of 1 percent of GDP lowers GDP by 0.62 percent within two years, and by 0.84 percent under pegged exchange rates (Guajardo et al., 2011). In the meta-regression of Gechert (2015), spending multipliers are close to one and 0.3 to 0.4 larger than tax and transfer multipliers. Alesina et al. (2018, 2019) find that a spending-based consolidation plan worth 1 percent of GDP lowers growth by about half a percentage point for less than two years, whereas a tax-based plan is followed by a fall in GDP of about 2 percent.

[6] A shift of 1 percentage point at $T + 10$ therefore changes growth by 0.44 points in 2029 (the first year after a four-year adjustment), by 1 point in 2034 and by 0.80 points in 2038. Over the ten post-adjustment years that the debt and deficit criteria evaluate, it averages 0.78 points, whereas during the adjustment years it lies between 0 and 0.33 points.

**Table 4:** Change in the SPB relative to the baseline: forecasts and model parameters, four-year adjustment (percentage points of GDP)

| | | Standalone criterion | | DSA target | | Binding target | |
|---|---|---|---|---|---|---|---|
| Input | Value | Mean | Largest | Mean | Largest | Mean | Largest |
| *1. Forecasts* | | | | | | | |
| Real short-term rate, $T + 10$ | −1 pp. | −0.05 | −0.15 ITA | −0.04 | −0.14 ITA | −0.04 | −0.14 ITA |
| | +1 pp. | +0.05 | +0.15 ITA | +0.04 | +0.14 ITA | +0.04 | +0.14 ITA |
| Real long-term rate, $T + 10$ | −1 pp. | −0.26 | −0.52 GRC | −0.23 | −0.52 GRC | −0.17 | −0.42 ESP |
| | +1 pp. | +0.29 | +0.53 GRC | +0.22 | +0.47 GRC | +0.18 | +0.40 ITA |
| Inflation, $T + 10$ | −0.5 pp. | +0.13 | +0.31 ITA | +0.10 | +0.30 ITA | +0.10 | +0.30 ITA |
| | +0.5 pp. | −0.11 | −0.30 ITA | −0.06 | −0.18 FIN | −0.06 | −0.12 BEL |
| Potential growth, $T + 10$ | −0.75 pp. | +0.46 | +0.91 ITA | +0.38 | +0.91 ITA | +0.33 | +0.91 ITA |
| | +0.75 pp. | −0.41 | −0.88 ITA | −0.28 | −0.49 ESP | −0.21 | −0.49 ESP |
| Ageing costs | −0.5 pp. | −0.22 | −0.36 BEL | −0.19 | −0.35 AUT | −0.13 | −0.34 BEL |
| | +0.5 pp. | +0.25 | +0.37 AUT | +0.18 | +0.35 AUT | +0.14 | +0.34 BEL |
| Real short-term rate, $T + 30$ | −0.5 pp. | 0.00 | −0.01 PRT | 0.00 | −0.01 PRT | 0.00 | −0.01 PRT |
| | +0.5 pp. | 0.00 | +0.01 PRT | 0.00 | +0.01 PRT | 0.00 | +0.01 PRT |
| Real long-term rate, $T + 30$ | −0.5 pp. | −0.01 | −0.03 GRC | −0.01 | −0.02 GRC | −0.01 | −0.02 GRC |
| | +0.5 pp. | +0.01 | +0.03 GRC | +0.01 | +0.02 GRC | +0.01 | +0.02 GRC |
| Inflation, $T + 30$ | −0.25 pp. | +0.01 | +0.04 GRC | +0.01 | +0.04 GRC | +0.01 | +0.03 BEL |
| | +0.25 pp. | −0.01 | −0.04 GRC | −0.01 | −0.04 GRC | −0.01 | −0.03 BEL |
| Ageing costs, previous vintage | 2021 Ageing Report | +0.05 | −1.44 ESP | +0.05 | −1.37 ESP | +0.07 | −1.37 ESP |
| Ageing costs, 2024 AR scenario | higher migration | −0.15 | −0.46 ESP | −0.13 | −0.43 ESP | −0.09 | −0.43 ESP |
| | lower fertility | −0.16 | −0.36 PRT | −0.15 | −0.34 PRT | −0.11 | −0.33 BEL |
| *2. Model parameters* | | | | | | | |
| Fiscal multiplier | 0.5 | −0.02 | −0.08 SVK | −0.08 | −0.32 POL | −0.26 | −1.42 FIN |
| | 0.6 | −0.01 | −0.05 SVK | −0.05 | −0.20 POL | −0.17 | −0.94 FIN |
| | 1 | +0.02 | +0.09 SVK | +0.12 | +0.64 FRA | +0.45 | +2.61 FIN |
| | 1.5 | +0.06 | +0.27 SVK | +0.63 | +2.89 FRA | $+1.35^{(12)}$ | +6.99 DEU |
| Multiplier persistence | 2 years | −0.02 | −0.07 FIN | −0.07 | −0.27 ROU | −0.22 | −1.35 FIN |
| | 4 years | +0.05 | +0.26 FIN | +0.07 | +0.22 FIN | +0.17 | +1.05 FIN |
| | 5 years | +0.10 | +0.49 FIN | +0.12 | +0.49 FIN | +0.30 | +1.88 FIN |
| Impulse-response output gap rule | 2 years | −0.02 | −0.09 SVK | −0.10 | −0.41 ROU | −0.29 | −1.82 FIN |
| | 3 years | −0.02 | −0.10 FIN | −0.06 | −0.28 ROU | −0.17 | −1.17 FIN |
| | 4 years | −0.01 | −0.13 FIN | 0.00 | +0.11 FRA | −0.02 | −0.48 FIN |
| | 5 years | +0.02 | +0.12 FRA | +0.05 | +0.33 FRA | +0.09 | +0.40 AUT |
| Budget semi-elasticity | ×0.9 | −0.01 | −0.05 FIN | −0.03 | −0.12 FRA | −0.10 | −0.53 FIN |
| | ×1.1 | +0.01 | +0.05 FIN | +0.03 | +0.16 FRA | +0.11 | +0.60 FIN |

Source: Author's calculations. Note: Mean: mean change across the 15 reference-trajectory countries with adjustment needs. Largest: largest change in absolute terms among the 15 countries, with its country. Standalone criterion: SPB required by the criterion on which the check acts directly (the baseline debt criterion; Section 2.3). Runs in which the debt sustainability safeguard cannot be met (a multiplier of 1.5 for Finland, Austria and Spain) are left out of the binding target, and a superscript in brackets gives the number of feasible countries.

Because these shifts phase in toward $T + 10$, they average about four-fifths of their full size over the ten post-adjustment evaluation years and reach at most a third of their size by 2028. Applying five-year-ahead WEO errors at $T + 10$, alongside German forward-rate errors that omit sovereign spread risk, therefore provides a lower bound on medium-term forecast uncertainty. Moreover, the debt sustainability safeguard and the EDP, which apply during the adjustment years, see almost no medium-term forecast error, just as the deterministic stress tests start only after the adjustment period ends (Wieland, 2024). Section 4.3 therefore adds a direct check of real growth in the first adjustment year.

## 3.3 Individual variation of inputs

Table 4 reports the mean and the largest changes in the required SPB across the three levels for a four-year adjustment. Table A3a in the Annex reports the seven-year results, where medium-term forecasts matter slightly more because the ten post-adjustment evaluation years lie closer to $T + 10$. Because an input moves a target only through the criterion that binds, the cross-country means hide large differences across member states: the DSA target responds most to lower potential growth in Italy, where the adverse $r - g$ test binds, to the ageing-cost assumptions in Spain, and to the fiscal multiplier in France and Poland, where the deficit criterion binds. Table A10 in the Annex lists for each country the criteria that set its targets and the three checks with the largest effects on its DSA target.

Among the forecasts, potential growth and the long-term interest rate have the largest effects. Potential growth 0.75 percentage points lower at $T + 10$ raises the average DSA target by 0.38 percentage points of GDP, with the largest increases in Italy (0.91 points), France (0.67) and Greece (0.67). Potential growth 0.75 points higher lowers the average DSA target by only 0.28 points, and by at most 0.49 points (in Spain). This asymmetry comes from the interaction of the criteria described in Section 2.3: on the standalone baseline debt criterion, the required SPB responds almost symmetrically (+0.46 versus −0.41 points), whereas for the DSA target, a favourable growth shift soon causes the deficit criterion to bind.

A real long-term interest rate 1 percentage point higher or lower at $T + 10$ moves the average DSA target almost symmetrically, by 0.22 and 0.23 points, changing the target by more than 0.1 points in 11 of the 15 countries and by up to 0.47 points in Greece. The short-term rate has almost no effect, and the long-run anchors at $T + 30$ change no target by more than 0.04 points. With real rates held constant, lower inflation mainly affects high-debt countries with long debt maturities: inflation 0.5 percentage points lower raises the DSA target by 0.30 points in Italy and 0.17 points in Greece (against 0.10 points on average) by slowing the nominal erosion of fixed-rate legacy debt. Conversely, higher inflation can raise a target when the deficit criterion binds, because with constant real rates it raises nominal interest rates and the interest bill, tightening the headline deficit criterion while easing the debt criteria.

A shift in ageing costs of 0.5 percentage points of GDP by 2038 moves the average DSA target by 0.18 to 0.19 points, and by up to 0.35 points in Austria. Shifts of this size are comparable to the demographic scenarios of the 2024 Ageing Report, which lower the projected 2024–2038 change in ageing costs by 0.28 points on average under higher migration and by 0.26 points under lower fertility. Replacing each country's baseline projection with their respective demographic scenarios lowers the average DSA target by 0.13 and 0.15 points, led by reductions of 0.43 points in Spain under higher migration and 0.34 points in Portugal under lower fertility. The effects remain modest because the four-year DSA horizon ends in 2038: up to that year, lower fertility mainly reduces education spending, whereas its upward pressure on pension ratios comes later. Over 2024 to 2070, the lower-fertility scenario raises ageing costs by 0.65 points on

average, while higher migration lowers them by 0.59 points, about twice its effect through 2038. Replacing the 2024 Ageing Report projections with those of the 2021 report shifts individual targets much more. Here, Spain's falls by 1.37 points (largely reflecting its 2021–2023 pension reforms), Greece's falls by 0.99 points, and Slovenia's rises by about 1 point (Section 6 examines the ten-year window over which these projections enter the DSA).

Within the typical range of published estimates, the model parameters change the average DSA target little. Raising the fiscal multiplier from 0.75 to 1.0 increases it by 0.12 percentage points (and by at most 0.64 points in France), while varying the persistence of the multiplier, the output gap rule or the budget semi-elasticity changes the average target by less than 0.15 points. Section 4 shows that the multiplier and the output gap rule matter much more for countries with binding safeguards.

## 3.4 Joint variation of forecasts and model parameters

Varying inputs one at a time misses their interactions, such as a multiplier that is both high and persistent or adverse forecasts that push several criteria above the baseline target at once. I therefore take 500 joint draws of all forecasts and model parameters using Latin hypercube sampling, which spreads the draws evenly over the marginal distribution of each input. For each forecast, the sampling selects a rank in the sorted historical errors and draws that observation together with its paired variable from the same dataset: short- and long-term rate errors come from the same Bund year, and growth and inflation errors come from the same country and WEO vintage. Because of incomplete sample overlap, I draw errors from separate datasets independently. Instead pairing WEO and Bund errors by calendar year yields nearly identical results, raising the average interquartile range of the binding target from 1.07 to 1.14 percentage points.[7] The fiscal multiplier is drawn uniformly from 0.5 to 1.0, the budget semi-elasticity scaling factor uniformly from 0.9 to 1.1, and multiplier persistence with equal probability from two, three and four years. In a given draw, all countries receive the same errors and parameters, while the estimation choices and calibration conventions stay at their baseline values.

Figure 2 and Table 5 summarise the country-level distributions by the median and the interquartile range across the 500 draws, keeping extreme historical errors, like the Eurozone crisis, from driving the headline results. For the 15 reference-trajectory countries with adjustment needs, the baseline binding target requires an average adjustment of 3.33 percentage points of GDP over four years. Across the draws, the 25th percentile of each country's binding target lies on average 0.39 points below the baseline, the median 0.07 points above it and the 75th percentile 0.68 points above it.[8] The interquartile range thus averages 1.07 points, about a third of

[7] Across the 302 overlapping observations from 2000 to 2015, pairing each WEO observation with the German forward-rate error in the same year and applying the rate errors directly to nominal interest rates yields a positive correlation of long-term rate errors with growth errors (0.27), which narrows the range of $r - g$, and a negative correlation with inflation errors (−0.40), which widens it; the average median of the binding target changes by −0.04 points.

[8] Cross-country statistics in this section report the mean of the country-level percentiles.

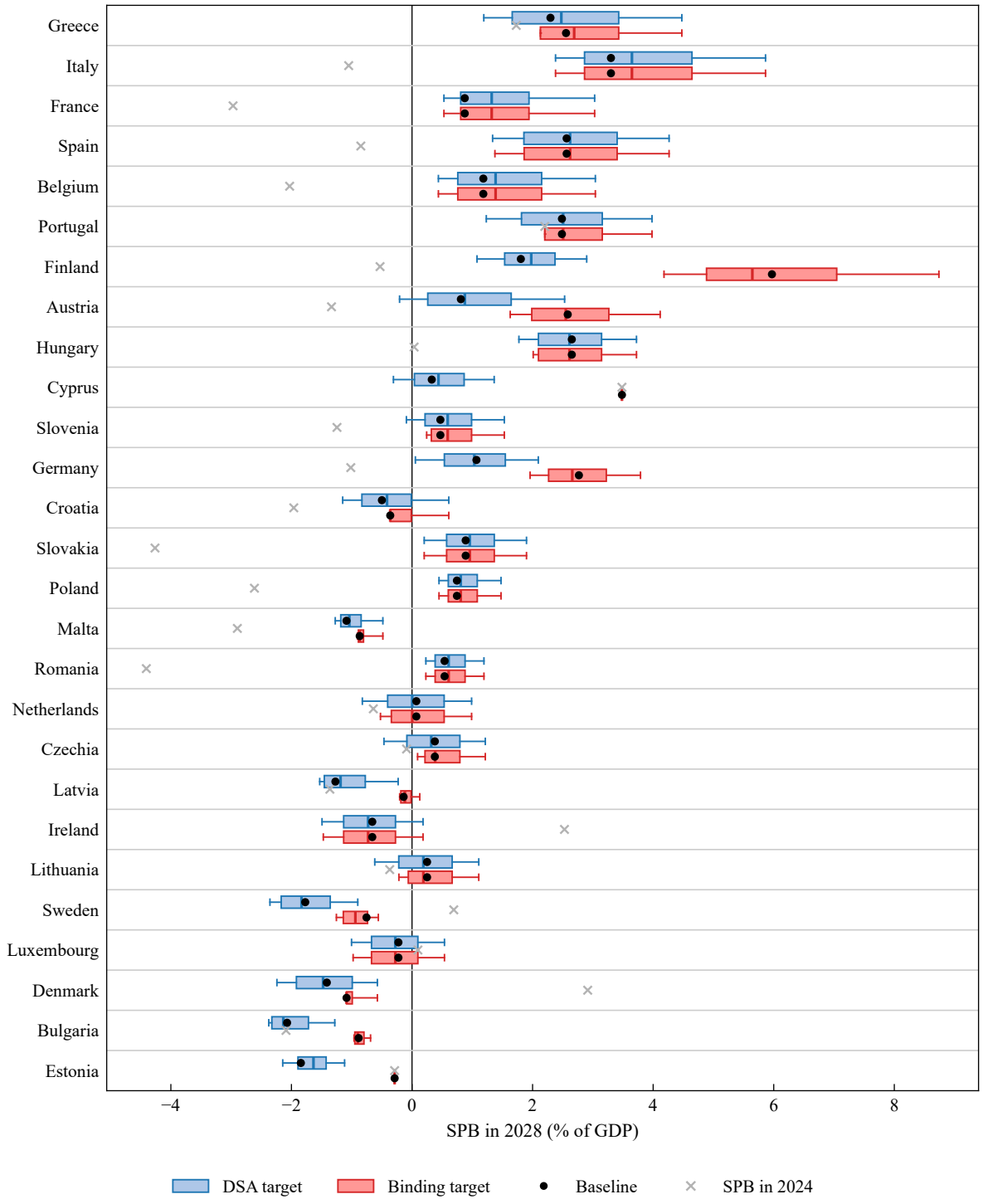



**Figure 2:** SPB targets by country: baseline and joint draws, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: SPB at the end of the adjustment period (2028). For each country, boxes show the 25th to 75th percentiles of the DSA target (blue) and of the binding target (red) across 500 joint draws of forecasts and model parameters, lines in the boxes mark the median and whiskers span the 10th to 90th percentiles. Dots show the baseline targets and grey crosses the SPB in 2024. For Cyprus and Estonia, the adjustment floor sets the binding target in 500 and 488 of the 500 draws, so the box collapses to the 2024 SPB. Countries are ordered by debt in 2024.

the required adjustment. The band is skewed upwards even for the standalone baseline debt cri-

**Table 5:** SPB targets by country: baseline and joint draws of forecasts and model parameters, four-year adjustment (percent of GDP and percentage points)

| | | | Standalone | DSA target | | | | | Binding target | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Debt | Required | Band | Base | P25 | Median | P75 | Band | Base | P25 | Median | P75 | Band |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) |
| Greece | 153.9 | 0.83 | 2.10 | 2.30 | −0.63 | +0.18 | +1.13 | 1.77 | 2.55 | −0.43 | +0.14 | +0.87 | 1.30 |
| Italy | 138.6 | 4.35 | 2.19 | 3.30 | −0.44 | +0.35 | +1.34 | 1.78 | 3.30 | −0.44 | +0.35 | +1.34 | 1.78 |
| France | 112.4 | 3.84 | 1.50 | 0.87 | −0.07 | +0.45 | +1.07 | 1.13 | 0.87 | −0.07 | +0.45 | +1.07 | 1.13 |
| Spain | 105.5 | 3.41 | 1.75 | 2.56 | −0.71 | +0.06 | +0.84 | 1.55 | 2.56 | −0.71 | +0.06 | +0.84 | 1.54 |
| Belgium | 103.4 | 3.21 | 1.85 | 1.18 | −0.42 | +0.20 | +0.97 | 1.39 | 1.18 | −0.42 | +0.20 | +0.97 | 1.39 |
| Portugal | 95.6 | 0.28 | 1.43 | 2.49 | −0.67 | +0.02 | +0.67 | 1.34 | 2.49 | −0.28 | +0.02 | +0.67 | 0.95 |
| Finland | 80.5 | 6.50 | 0.83 | 1.81 | −0.27 | +0.17 | +0.57 | 0.84 | 5.97 | −1.09 | −0.33 | +1.07 | 2.16 |
| Austria | 79.5 | 3.92 | 1.55 | 0.81 | −0.55 | +0.07 | +0.83 | 1.38 | 2.58 | −0.60 | −0.03 | +0.68 | 1.28 |
| Hungary | 74.3 | 2.62 | 1.26 | 2.65 | −0.55 | −0.04 | +0.49 | 1.05 | 2.65 | −0.55 | −0.04 | +0.49 | 1.05 |
| Cyprus | 70.6 | 0.00 | 0.89 | 0.33 | −0.29 | +0.11 | +0.54 | 0.82 | 3.48 | 0.00 | 0.00 | 0.00 | 0.00 |
| Slovenia | 68.1 | 1.72 | 0.86 | 0.47 | −0.25 | +0.12 | +0.52 | 0.77 | 0.47 | −0.15 | +0.12 | +0.52 | 0.67 |
| Germany | 62.5 | 3.78 | 0.93 | 1.07 | −0.53 | −0.04 | +0.48 | 1.01 | 2.77 | −0.50 | −0.11 | +0.45 | 0.96 |
| Croatia | 59.5 | 1.60 | 0.88 | −0.50 | −0.33 | +0.08 | +0.49 | 0.82 | −0.36 | 0.00 | 0.00 | +0.35 | 0.35 |
| Slovakia | 58.5 | 5.15 | 0.83 | 0.89 | −0.32 | +0.07 | +0.47 | 0.79 | 0.89 | −0.32 | +0.07 | +0.47 | 0.79 |
| Poland | 53.7 | 3.36 | 0.91 | 0.74 | −0.14 | +0.06 | +0.34 | 0.48 | 0.74 | −0.14 | +0.06 | +0.34 | 0.48 |
| Malta | 52.0 | 2.03 | 0.84 | −1.09 | −0.10 | +0.05 | +0.24 | 0.34 | −0.87 | −0.02 | 0.00 | +0.06 | 0.09 |
| Romania | 50.9 | 4.95 | 0.83 | 0.54 | −0.16 | +0.07 | +0.34 | 0.50 | 0.54 | −0.16 | +0.07 | +0.34 | 0.50 |
| Netherlands | 47.1 | 0.71 | 0.81 | 0.07 | −0.48 | −0.07 | +0.46 | 0.94 | 0.07 | −0.41 | −0.07 | +0.46 | 0.88 |
| Czechia | 45.2 | 0.47 | 0.77 | 0.38 | −0.46 | −0.06 | +0.42 | 0.88 | 0.38 | −0.16 | +0.01 | +0.42 | 0.58 |
| Latvia | 44.5 | 1.22 | 0.81 | −1.27 | −0.18 | +0.09 | +0.50 | 0.68 | −0.14 | −0.05 | +0.02 | +0.12 | 0.17 |
| Ireland | 42.5 | −3.19 | 0.77 | −0.66 | −0.47 | −0.07 | +0.39 | 0.86 | −0.66 | −0.47 | −0.07 | +0.39 | 0.86 |
| Lithuania | 38.3 | 0.62 | 0.76 | 0.25 | −0.47 | −0.06 | +0.42 | 0.89 | 0.25 | −0.31 | −0.06 | +0.42 | 0.73 |
| Sweden | 32.0 | −1.45 | 0.77 | −1.77 | −0.40 | −0.06 | +0.41 | 0.81 | −0.75 | −0.38 | −0.18 | +0.02 | 0.40 |
| Luxembourg | 27.1 | −0.32 | 0.73 | −0.23 | −0.45 | −0.05 | +0.32 | 0.77 | −0.23 | −0.45 | −0.05 | +0.32 | 0.77 |
| Denmark | 26.5 | −4.00 | 0.81 | −1.41 | −0.50 | −0.06 | +0.42 | 0.93 | −1.09 | 0.00 | 0.00 | +0.09 | 0.09 |
| Bulgaria | 24.5 | 1.20 | 0.69 | −2.07 | −0.25 | −0.06 | +0.35 | 0.61 | −0.89 | −0.06 | −0.02 | +0.09 | 0.15 |
| Estonia | 21.4 | 0.00 | 0.72 | −1.84 | −0.05 | +0.21 | +0.42 | 0.47 | −0.29 | 0.00 | 0.00 | 0.00 | 0.00 |
| Reference trajectory (15) | 86.0 | 3.33 | 1.31 | 1.37 | −0.39 | +0.12 | +0.69 | 1.07 | 1.91 | −0.39 | +0.07 | +0.68 | 1.07 |
| All with adjustment needs (21) | 73.7 | 2.66 | 1.16 | 0.83 | −0.38 | +0.08 | +0.62 | 1.00 | 1.33 | −0.33 | +0.04 | +0.57 | 0.90 |
| Technical information (6) | 43.2 | 0.97 | 0.79 | −0.52 | −0.36 | −0.01 | +0.44 | 0.80 | −0.11 | −0.17 | −0.02 | +0.31 | 0.48 |

Source: Author's calculations. Note: Column (1) is debt in 2024. Column (2) is the adjustment required by the baseline binding target relative to the 2024 SPB. Column (3) is the interquartile band of the SPB required by the standalone criterion (the baseline deterministic debt criterion). Base is the baseline target. P25, Median and P75 are the 25th, 50th and 75th percentiles of the change in the target across 500 joint draws relative to the baseline; Band is P75 minus P25. The last three rows report the mean across the 15 reference-trajectory countries with adjustment needs, across all 21 countries with adjustment needs and across the six technical-information countries with adjustment needs.

terion because the historical distributions of growth and interest-rate errors have longer adverse tails (Table A1 in the Annex). For the DSA target, the 25th and 75th percentiles lie 0.39 points below and 0.69 points above the baseline, whereas the targets of the six technical-information countries vary less because they face neither the stress tests nor the stochastic criterion.

The interquartile range represents both the span of targets that a government could defend using inputs inside historical error ranges—or that can arise across successive planning cycles as forecasts are revised—and the band of forecast uncertainty around the sustainable SPB. For countries with small baseline adjustments, this range exceeds the required adjustment itself: in Greece, the interquartile band of the binding target is 1.30 points against a required adjustment of 0.83 points, and in Portugal it is 0.95 points against 0.28 points. Across countries, the band of the DSA target is widest for Italy and Greece, at 1.78 and 1.77 percentage points. Because all countries draw from the same pooled errors and the same German yield curve, these differences reflect how sensitive each country's target is to a given error rather than country-specific forecast uncertainty. The asymmetry between favourable and adverse draws is largest in France, where

the deficit criterion sets the DSA target in 49 percent of the draws and the adverse $r - g$ test in the remainder: because favourable draws quickly hit the competing criterion, France's 25th percentile lies just 0.07 points below the baseline, while its median and 75th percentile lie 0.45 and 1.07 points above it.

The debt sustainability safeguard sets the binding target in most draws for Finland (100 percent), Germany (98 percent) and Austria (90 percent). It widens the band relative to the DSA target in Finland (where the 75th percentile rises from 0.57 to 1.07 points above the baseline), but narrows it slightly in Austria (1.28 versus 1.38 points) and Germany (0.96 versus 1.01 points) because the safeguard evaluates debt only during the adjustment years, when the $T + 10$ forecast shifts have barely begun; as Section 4.3 shows, however, the safeguard responds strongly to near-term growth errors that hit during the adjustment window itself.

Across the 15 countries and 500 draws, 21 percent of country-draws require an adjustment of more than 5 percentage points of GDP over four years, a magnitude observed in fewer than 10 percent of consolidation episodes of up to seven years across 36 advanced economies between 1970 and 2022 (Darvas et al., 2025a). This share reaches 85 percent in Finland and 62 percent in Slovakia, where the baseline already requires an adjustment above 5 points. In the actual application of the rules, SPB paths are of course agreed ex-ante, which is why realised forecast errors do not affect SPB targets but rather the debt trajectory itself. Projecting each country's debt ratio under the baseline SPB path across the 500 draws results in an interquartile range of the debt ratio of 12.0 percentage points of GDP on average for the 15 countries (Figure A1 in the Annex).

# 4 Fiscal multiplier and debt sustainability safeguard

Within the typical range of published estimates, the fiscal multiplier barely alters the core DSA targets (Section 3.3). However, it matters much more for the two constraints defined in headline terms, the deficit criterion and the debt sustainability safeguard. This is because a larger multiplier lowers output and with it the cyclical budget balance. The safeguard is the more sensitive of the two constraints, as it tests the decline in debt during the adjustment period, when adjustment-driven output losses peak.

## 4.1 Headline constraints

A larger multiplier widens the negative output gap opened by the adjustment. In the Commission's calculation sheets, the output gap follows a recursive rule, where two-thirds of the previous year's output gap deviation carries over to the current year. This causes the output losses of earlier steps to accumulate during the adjustment (see Annex A.1 for details). A wider output gap lowers nominal GDP and the headline balance while leaving the structural balance, and thus the deficit resilience safeguard, unchanged.

As a first-order approximation, a simple impact calculation illustrates this effect on the headline balance by abstracting from interest rates, inflation and lagged output dynamics (see also Boussard et al., 2012; Eyraud and Weber, 2013). Let $m$ denote the fiscal multiplier and $\varepsilon$ the budget semi-elasticity. An increase in the SPB by $x$ percentage points improves the headline balance on impact by $x(1 - \varepsilon m)$. For a semi-elasticity of 0.5, the headline balance improves by $0.63x$ at the baseline multiplier of 0.75, but by only $0.25x$ at a multiplier of 1.5.

This explains why a high multiplier sharply raises the DSA targets set by the deficit criterion. Under the crisis multiplier of 1.5, the average DSA target rises by 0.63 points, led by increases in France (2.89 points), Poland (1.57), Italy (1.36), Belgium (1.24) and Romania (1.15). At this value, the multiplier has the largest effect of all checks on the DSA target in Belgium, France, Italy, Malta, Poland and Romania (Table A10).

## 4.2 The multiplier and the safeguard

The debt sustainability safeguard requires the debt ratio to fall during the adjustment period itself, when a larger multiplier lowers nominal GDP (Heimberger et al., 2024). Let $b$ denote the initial debt ratio as a share of GDP. To first order, an SPB adjustment of $x$ points lowers the debt ratio by $x(1 - \varepsilon m)$ on impact through the smaller deficit, but raises it by about $bmx$ through lower GDP in the denominator. The debt ratio therefore falls on impact only if $m(b + \varepsilon) < 1$. For a debt ratio of 80 percent of GDP and a semi-elasticity of 0.5, this requires a multiplier below 0.77. Over several years, the safeguard can be met at somewhat higher multipliers because the output effect fades while the lower deficit continues to reduce debt, yet at the baseline multiplier the impact condition already fails for every country with debt above 83 percent of GDP. The safeguard does not bind in Italy, France or Belgium only because they remain in an EDP throughout the four adjustment years (where the safeguard does not apply), while in Greece, Spain and Portugal debt falls fast enough in the baseline.

The binding target is therefore highly sensitive to the multiplier and the output gap rule in the countries where the safeguard binds.[9] Raising the multiplier from 0.75 to 1.0 raises the binding target by 2.61 percentage points in Finland, 1.48 points in Austria and 1.00 point in Germany, and a persistence of five years raises Finland's by 1.88 points. With a crisis multiplier of 1.5, Germany's binding target rises by 6.99 points, and Finland, Austria and Spain cannot meet the safeguard with any annual adjustment up to 5 percentage points. Conversely, the Commission's 2026 multiplier of 0.6 lowers the binding targets of Finland, Austria and Germany by 0.94, 0.55 and 0.42 points, while replacing the Commission's recursive output gap rule with the linear

[9] How the safeguard is computed also matters. Like the Commission's sheets, I assume a constant annual adjustment, whereas the rules also permit a front-loaded path: with annual steps that decline linearly to zero, the four-year binding targets of Finland, Austria and Germany fall from 5.97 to 3.56, from 2.58 to 1.16 and from 2.77 to 1.50 percent of GDP. Similarly, our reading of Article 7(2) of Regulation (EU) 2024/1263 in Darvas et al. (2024) measures the average decline in debt across the whole adjustment period from the debt ratio in year $T$, which gives the same targets on the 2024 data but diverges on the 2026 data (Table A7). The accompanying repository compares these implementations.

impulse-response rule of Annex A.1 lowers Finland's baseline binding target by 1.17 points and cuts the effect of a 1.5 multiplier in Germany from 6.99 to 4.07 points (Table A8 in the Annex).

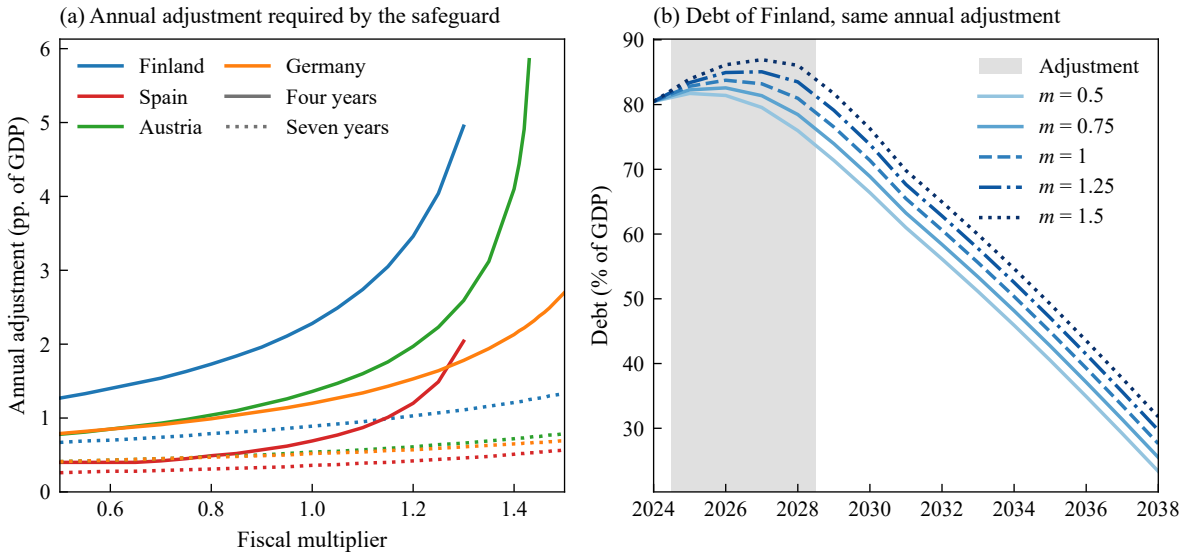



**Figure 3:** Debt sustainability safeguard and fiscal multiplier

Source: Author's calculations. Note: (a) Average annual adjustment of the path with the smallest constant annual adjustment, in steps of 0.01 percentage points, for which the debt sustainability safeguard is met (including the minimum steps of the EDP and the deficit resilience safeguard) under the Commission's output gap rule, for multipliers from 0.5 to 1.5. Lines end where no annual adjustment up to 20 percentage points meets the safeguard. Solid lines: four-year adjustment; dotted lines: seven-year adjustment. (b) Debt of Finland for a constant annual adjustment of 1.63 percentage points (its baseline four-year binding adjustment).

Figure 3 shows how the multiplier alters the required adjustment and the path of debt. Panel (a) plots the smallest constant annual adjustment that meets the safeguard for the four countries where the safeguard binds or becomes infeasible. Over four years, it ranges from 0.45 percentage points per year in Spain to 1.63 points in Finland at the baseline multiplier, rising gradually up to a multiplier of about 1.2 and steeply above it; no annual adjustment up to 20 points meets the safeguard in Finland and Spain above a multiplier of 1.30, or in Austria above 1.43. Panel (b) shows that a higher multiplier mainly delays the decline in debt. Under Finland's baseline annual adjustment of 1.63 points, debt falls from 80.5 percent of GDP in 2024 to 76.0 percent in 2028 with a multiplier of 0.5, but rises to 86.1 percent with a multiplier of 1.5, missing the safeguard for all multipliers of 1.0 or higher. Once the adjustment ends and the output gap closes, however, debt falls by about 5.3 points per year regardless of the multiplier, reaching 23.3 and 31.8 percent of GDP in 2038 at the two extremes.

## 4.3 Forecast errors during the adjustment

Forecast errors that lower output during the adjustment act on the safeguard in the same way as a higher multiplier. To show this, I shift real GDP growth in 2025 (the first adjustment year) by ±1.25 percentage points, about half the demeaned interquartile range of one-year-ahead WEO

real growth errors (−1.07 to +1.47 points). Holding potential growth fixed causes a reaction in the output gap closure. Growth 1.25 points lower in 2025 raises the binding target by 1.45 points in Finland, 1.41 points in Austria and 1.11 points in Germany, while higher growth lowers them symmetrically; across all other countries, the binding target changes by at most 0.40 points (Table A9 in the Annex). In Austria and Germany, a one-year growth shortfall of this size thus moves the binding target by about as much as raising the multiplier from 0.75 to 1.0 (1.41 versus 1.48 points in Austria, and 1.11 versus 1.00 points in Germany).

Because medium-term plans fix a path for net expenditure, neither a higher multiplier nor a cyclical growth shortfall raises the required adjustment after the fact. Instead, debt simply falls less during the adjustment than the safeguard assumed on paper. As Figure 3b shows, this temporary shortfall has no bearing on medium-term debt sustainability, as debt trajectories converge once output recovers. However, enforcing rigid debt reduction during the consolidation window penalises projected cyclical downturns, revealing a fundamental disconnect between the legal design of annual safeguards and economic reality. Seven-year plans, which Finland, Austria, Germany and Spain obtained, roughly halve the effect of a first-year growth shortfall on the binding targets of the first three (Table A9) and remove the infeasibility under high multipliers, allowing the safeguard to be met with annual adjustments of at most 1.34 percentage points (dotted lines in Figure 3a).

# 5 Estimation choices in the stochastic analysis

The third group of inputs contains the estimation choices of the stochastic criterion. To show how these choices affect SPB targets, I vary them individually. These choices include the chosen model for the estimation of shocks, the start of the shock sample, the frequency of the underlying historical data, and their winsorisation (Table 6). For the estimation model, I compare the baseline normal draws with the Commission's 2026 method, which filters each shock series with a univariate autoregressive model and resamples the residuals (the AR bootstrap), and with a six-variable vector autoregression (VAR) bootstrap as in Darvas et al. (2025a). I evaluate both bootstrap methods with and without the inclusion of the estimated autoregressive intercepts (Annex A.2).

While these choices shift the SPB required by the standalone stochastic criterion, almost none of the variation passes through to the DSA targets because the stochastic criterion binds only for Finland in the baseline and lies well below the deterministic criteria in most other countries. Across the 15 countries, winsorising the shocks at the 1st and 99th percentiles instead of the 5th and 95th raises the SPB required by the standalone stochastic criterion by 0.53 percentage points on average, removing winsorisation raises it by 0.84 points, and using annual instead of quarterly shocks lowers it by 0.15 points. However, only winsorisation moves the overall DSA targets materially. Relaxing it to the 1st and 99th percentiles raises the DSA target in Finland (0.27 points), France (0.26) and Austria (0.25), while removing winsorisation raises the targets

**Table 6:** Change in the SPB relative to the baseline: estimation choices, calibration conventions and data, four-year adjustment (percentage points of GDP)

| | | Standalone criterion | | DSA target | | Binding target | |
|---|---|---|---|---|---|---|---|
| Input | Value | Mean | Largest | Mean | Largest | Mean | Largest |
| *3. Estimation choices* | | | | | | | |
| Start of shock sample | 1990 | −0.01 | −0.03 FRA | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2010 | +0.02 | +0.38 SVN | 0.00 | +0.07 FIN | 0.00 | 0.00 |
| Shock frequency | annual | −0.15 | −0.57 MLT | +0.01 | +0.09 FIN | 0.00 | 0.00 |
| Shock model | VAR bootstrap | −0.16 | −0.52 MLT | 0.00 | −0.02 FIN | 0.00 | 0.00 |
| | AR bootstrap | −0.26 | −0.60 MLT | −0.01 | −0.08 FIN | 0.00 | 0.00 |
| | VAR bootstrap, intercept | +0.05 | −0.69 ROU | +0.03 | +0.45 FIN | 0.00 | 0.00 |
| | AR bootstrap, intercept | −0.02 | −0.66 MLT | +0.03 | +0.46 FIN | 0.00 | 0.00 |
| | AR bootstrap, not winsorised | +0.13 | +0.64 ITA | +0.01 | +0.10 FIN | 0.00 | 0.00 |
| Winsorised shocks | P1 and P99 | +0.53 | +1.17 ESP | +0.05 | +0.27 FIN | +0.02 | +0.26 FRA |
| | off | +0.84 | +1.73 ESP | +0.15 | +0.66 AUT | +0.07 | +0.57 FRA |
| *4. Calibration conventions* | | | | | | | |
| Start of stochastic projection | first adjustment year | +0.19 | +0.81 GRC | 0.00 | +0.06 FIN | 0.00 | 0.00 |
| Horizon of stochastic projection | 10 years | +0.10 | −0.68 FIN | −0.01 | −0.08 FIN | 0.00 | 0.00 |
| Threshold, start, horizon | 60%, after adj., 5 years | −0.50 | −0.91 GRC | −0.01 | −0.08 FIN | 0.00 | 0.00 |
| | 80%, after adj., 5 years | +0.57 | +1.00 GRC | +0.05 | +0.43 FIN | +0.01 | +0.06 GRC |
| | 80%, first adj. year, 5 years | +0.92 | +2.46 GRC | +0.21 | +1.55 GRC | +0.14 | +1.29 GRC |
| | 80%, first adj. year, 10 years | +0.87 | +2.96 GRC | +0.19 | +2.05 GRC | +0.16 | +1.79 GRC |
| Adverse $r - g$ shock | 0.25 pp. | −0.29 | −0.50 ITA | −0.08 | −0.28 GRC | −0.06 | −0.26 BEL |
| | 0.75 pp. | +0.28 | +0.50 ITA | +0.22 | +0.50 ITA | +0.17 | +0.50 ITA |
| Lower-SPB shock | 0.25 pp. | −0.23 | −0.25 AUT | −0.01 | −0.09 SVK | −0.01 | −0.09 SVK |
| | 0.75 pp. | +0.23 | +0.25 AUT | +0.09 | +0.24 HUN | +0.06 | +0.24 HUN |
| Financial stress shock | 0.5 pp. | −0.03 | −0.11 FRA | 0.00 | 0.00 | 0.00 | 0.00 |
| | 1.5 pp. | +0.03 | +0.11 FRA | 0.00 | 0.00 | 0.00 | 0.00 |
| Ageing cost period | 0 years | −0.65 | −1.96 PRT | −0.59 | −1.83 ESP | −0.34 | −1.62 ESP |
| | 5 years | −0.27 | −0.93 ESP | −0.24 | −0.89 ESP | −0.18 | −0.89 ESP |
| TFP growth after $T + 10$ | held at $T + 10$ value | +0.15 | +0.49 GRC | +0.15 | +0.44 GRC | +0.13 | +0.44 ESP |
| All settings | lenient | −0.27 | −0.93 ESP | −0.45 | −1.26 ESP | −0.27 | −1.26 ESP |
| | strict | +0.15 | +0.49 GRC | +0.41 | +0.86 GRC | +0.32 | +0.77 ITA |
| Threshold and stress tests | lenient | – | – | −0.19 | −0.34 ESP | −0.12 | −0.34 ESP |
| | strict | – | – | +0.25 | +0.50 ITA | +0.18 | +0.50 ITA |
| *Data* | | | | | | | |
| Repayment profile | Eurostat profile | −0.01 | −0.46 GRC | −0.02 | −0.57 GRC | +0.01 | −0.42 GRC |

Source: Author's calculations. Note: See Table 4. The standalone criterion of the stochastic settings and stress tests applies to the 15 reference-trajectory countries with adjustment needs. AR bootstrap: method of the Commission since the Debt Sustainability Monitor 2025 (Annex A.2); Intercept: retains the constants of the autoregressive models in the simulation. Threshold, start, horizon: probability threshold of the stochastic criterion (baseline 70 percent), start of the stochastic projection (after the adjustment period versus first adjustment year) and horizon after the adjustment period (five versus ten years); Table A4 in the Annex reports the other five combinations. Baseline stress tests apply shocks of 0.5, 1.0 and 0.5 percentage points in the adverse $r - g$, financial stress and lower-SPB tests, with a ten-year ageing-cost period. All settings, lenient: 60 percent threshold, shocks of 0.25, 0.5 and 0.25 points and a five-year ageing period. All settings, strict: 80 percent threshold, shocks of 0.75, 1.5 and 0.75 points and TFP growth held at its $T + 10$ value. Threshold and stress tests: the same risk-margin bundles without changing the ageing period or TFP growth (standalone criterion omitted for multi-criterion bundles). Hungary, Luxembourg and the Netherlands are omitted from the repayment profile check due to missing data.

of Austria and France by 0.66 and 0.57 points and that of Finland by 0.38 points. This is more than raising the probability threshold to 80 percent (at most +0.43 points) or increasing the lower-SPB shock (at most +0.24 points). Gros and Hofer (2026) similarly find that the stochastic criterion rarely binds because winsorisation removes the largest shocks, and Cottarelli (2024) objects to truncating these outliers since tail crises do occur in practice. I regard winsorisation as economically defensible because it prevents historical volatility from past structural breaks, such as the acute phase of the euro-area debt crisis, from permanently penalising countries

whose future risk profiles have normalised.

The AR bootstrap lowers the SPB required by the standalone stochastic criterion by 0.26 points on average, and the VAR bootstrap lowers it by 0.16 points, with neither changing any country's DSA target by more than 0.1 points (Finland's falls by 0.08 points under the AR bootstrap). This matches the Commission's assessment in the Debt Sustainability Monitor 2025 (European Commission, 2026, Box 2.3), where the 2026 method narrows the five-year debt cone (the 90th minus the 10th percentile) in every member state except Spain. Bec et al. (2025) trace this narrowing mainly to the mean-reverting persistence of shocks to the primary balance. Retaining the intercepts of the autoregressive models, as a literal reading of Annex A4.3 of European Commission (2026) and Bec et al. (2025) could imply, adds historical sample trends in first differences on top of the baseline projection, causing the simulated shocks to drift and raising Finland's DSA target by 0.46 points (see Annex A.2).

# 6 Calibration conventions

The fourth group of assumptions in the DSA contains settings that reflect normative choices about how much risk the rules accept and the judgement of projected long-run trends. Table 6 reports how the required SPB changes when changing these settings, both one at a time and in bundled scenarios.

A probability threshold of 80 instead of 70 percent raises the SPB required by the standalone stochastic criterion by 0.57 percentage points on average, and a threshold of 60 percent lowers it by 0.50 points, but an 80 percent threshold raises the average DSA target by only 0.05 points and changes only Finland's target by more than 0.1 points. Starting the stochastic shocks in the first adjustment year, as proposed in Darvas et al. (2023) (see also Wieland (2024); Erce (2025)), has little effect on its own while the criterion still compares debt five years after the adjustment with debt at its end. Combined with an 80 percent threshold, however, the early start raises the average DSA target by 0.21 points, driven by increases of 1.55 points in Greece and 0.55 points in Finland; adding a ten-year horizon on top raises Greece's target by 2.05 points.

Among the stress tests, the size of the adverse $r - g$ test matters most because it sets the DSA target in seven countries. Raising its interest-rate and growth shocks from 0.5 to 0.75 percentage points raises the standalone stress-test SPB requirement by 0.28 points on average, lifting the average DSA target by 0.22 points and Italy's by 0.50 points (while shocks of 0.25 points lower the average DSA target by 0.08 points). A lower-SPB shock of 0.75 points raises the average DSA target by only 0.09 points, and the size of the financial stress test changes no target.

Although the Commission motivated the permanent 1-point increase in $r - g$ qualitatively when introducing the test in the Fiscal Sustainability Report 2021 (European Commission, 2022) and applies the same shock across all countries regardless of their historical volatility (Cottarelli, 2024), the forecast errors of Section 3.1 provide an empirical benchmark. Across all pairwise combinations of the demeaned ten-year Bund forward-rate errors and five-year WEO

potential growth errors, the error in $r - g$ exceeded 1 percentage point in 30 percent of cases. Measured this way (and abstracting from spread errors), the baseline test covers about 70 percent of past permanent errors in $r - g$, and shocks of 0.75 points (a 1.5-point rise in $r - g$) cover about 77 percent. While the stochastic criterion also uses a 70 percent threshold, its shocks are country-specific, temporary, partly cancel out and are evaluated over five years rather than ten. The permanent adverse $r - g$ test therefore sets a higher bar and requires a higher SPB than the stochastic criterion in 12 of the 15 countries.

Alongside the uncertainty around ageing projections examined in Section 3.3, the window over which the rules pre-fund these costs moves the targets of most member states. Because the SPB before ageing costs is held constant after the adjustment period, any projected increase in public spending on pensions, health care, long-term care and education over the following ten years lowers the primary balance one-for-one and must be covered upfront through a higher SPB target (Darvas et al., 2025b). This ten-year window assumes that governments take no later tax or spending measures to offset demographic pressures. Shortening the window to five years lowers the average DSA target by 0.24 percentage points (and Spain's by 0.89 points), while raising the target slightly in Bulgaria and Latvia where ageing costs fall between the fifth and tenth years after the adjustment. Dropping the post-adjustment window altogether lowers the average DSA target by 0.59 points, and Spain's by 1.83 points. In 11 of the 21 countries with adjustment needs, either the ageing-cost window or the 2021 Ageing Report projections have the largest effect of all checks on the DSA target (Table A10).

A related long-run convention governs potential growth after $T + 10$, where the Ageing Report assumes that TFP growth converges to a common long-run rate, raising projected growth in countries with weak current TFP growth.[10] Holding TFP growth at its $T + 10$ level raises the average DSA target by 0.15 points, and Greece's by 0.44 points.

Figure 4 compares the effects of the four groups of inputs and evaluates two scenarios that bundle the calibration conventions (reported by country in Table A5 in the Annex). Because the baseline already sits at the strict boundary on ageing costs (covering the full ten-year horizon) and the lenient boundary on productivity (assuming full TFP convergence), the lenient scenario shortens the ageing window while the strict scenario turns off TFP convergence. Combining a 60 percent threshold, halved stress-test shocks and a five-year ageing window, the lenient scenario lowers the average DSA target by 0.45 points, slightly more than the 25th percentile of the joint draws (−0.39 points); most of this comes from the shorter ageing window, as the threshold and stress tests alone lower the target by only 0.19 points. In the strict scenario, an 80 percent threshold and 1.5× stress-test shocks alone raise the average DSA target by 0.25 points (about a third of the 0.69-point increase at the 75th percentile of the joint draws), and holding TFP growth at its $T + 10$ level brings the total increase to 0.41 points. The ageing window thus

[10] In March 2026, the Economic Policy Committee endorsed an extension of the commonly agreed potential output method from $T + 10$ to $T + 50$ (Arevalo et al., 2026), under which TFP growth converges to the EU rate by $T + 30$ for countries below the EU average and by $T + 55$ for countries above it. From the 2027 Ageing Report onward, the growth path after $T + 10$ will follow this method.

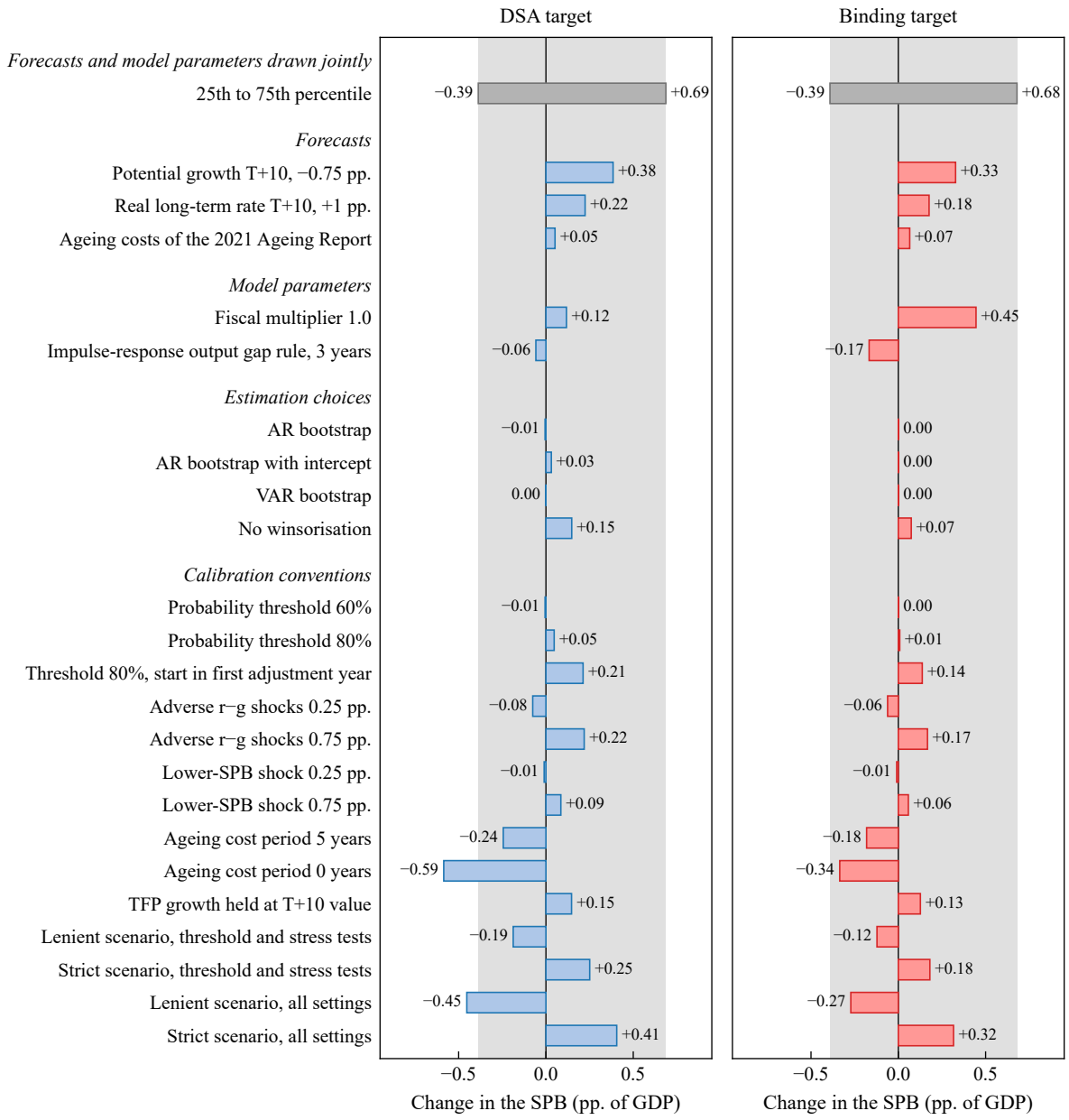



**Figure 4:** Forecasts, model parameters, estimation choices and calibration conventions: mean change in the targets, four-year adjustment (percentage points of GDP)

Source: Author's calculations. Note: Mean change in the target relative to the baseline across the 15 reference-trajectory countries with adjustment needs. Divide by four for the change in the annual adjustment. Grey bars and shading: means of the country-level 25th and 75th percentiles across the 500 joint draws of forecasts and model parameters. The other bars are individual checks and bundled settings from Tables 4 and 6. Runs in which the debt sustainability safeguard cannot be met are left out of the binding target.

moves the DSA targets by amounts comparable to forecast errors, the explicit risk margins move them less, and the specification of the stochastic fan chart hardly at all, except for winsorisation.

# 7 Robustness

## 7.1 Repayment profile of debt

Because the DSA criteria evaluate debt up to 14 years ahead, a large share of initial debt matures and is refinanced at market rates over the projection horizon. When market rates exceed the average interest rate on existing debt, the repayment profile determines how fast the implicit interest rate rises. In the baseline sheets, the share of long-term debt maturing each year starts from the share maturing in the latest year and converges to a six-year historical average (or follows official loan schedules in Greece). Applying a single rollover share to the entire stock can misstate refinancing speed in either direction: if a government shortens the maturity of new issuance when rates rise, applying the higher recent rollover share to the whole stock implies that legacy debt fixed at low rates is refinanced earlier than its contractual schedule requires. To separate existing debt from new issuance, the alternative check repays debt outstanding in 2024 according to Eurostat data on government debt by residual maturity (available for all member states except Hungary, Luxembourg and the Netherlands), while repaying newly issued long-term debt evenly over 20 years.

The check changes the targets mainly for Greece and Finland. For Greece, the Commission's inputs assume that the share of long-term debt maturing each year rises from about 4 percent in 2026 to between 7 and 11 percent after 2033, whereas in the Eurostat data this share stays between 2 and 4 percent because much of Greek debt consists of long-maturity loans from the European stability mechanisms. Because less debt is refinanced at market rates under the Eurostat profile, Greece's implicit interest rate in 2038 is 3.0 instead of 3.5 percent for a given SPB path, lowering 2038 debt by 3 percentage points of GDP and reducing Greece's DSA and binding targets by 0.57 and 0.42 points. For Finland, the effect runs the other way: based on ECB data that may refer to a narrower debt stock than full Maastricht debt, the Commission's inputs set the annual rollover share at 3 to 4 percent, whereas Eurostat's data on Finnish long-term Maastricht debt show about 14 percent maturing each year until 2028 and an average residual maturity of 7.8 years. Faster refinancing at higher market rates raises Finland's average interest rate in 2038 from 2.2 to 2.7 percent and its debt ratio by 3 points, increasing its DSA target by 0.27 points and its binding target by 0.35 points. Elsewhere, the DSA targets of Austria and Italy fall by 0.14 and 0.11 points, and no other country's target changes by more than 0.1 points.

## 7.2 Data of October 2026

To check whether the findings are specific to the 2024 data and method, I rerun all individual checks of the forecasts, parameters and calibration conventions using input data from October 2026 alongside the Commission's 2026 method (Annex A.4). The reference year is then 2026, the four-year adjustment runs from 2027 to 2030, the fiscal multiplier is 0.6 and the stochastic

shocks follow the AR bootstrap (under which the stochastic criterion binds for no country).[11] While baseline targets shift between the two data vintages (Table A7 in the Annex), the average sensitivity to each input stays very close to the 2024 results: across all checks in Table A6 in the Annex, the mean effects with the 2026 data and method differ from those with the 2024 data by 0.03 percentage points on average for the DSA target (at most 0.14 points, for shocks without winsorisation) and by 0.05 points for the binding target. For the binding target, a multiplier of 1.0 raises the average target by 0.36 points more with the 2026 data, and at a multiplier of 1.5—where the set of countries in which the debt sustainability safeguard becomes infeasible shifts between vintages—the effect across the 11 countries that remain feasible in both vintages is 0.95 points larger with the 2026 data. Because the baseline multiplier of the 2026 method is lower (0.6 versus 0.75), the step up to 1.0 or 1.5 is larger, with the extra effect concentrated in Finland, Germany and Slovenia.

These stable cross-country averages mask shifts for individual countries as updated fiscal and macroeconomic conditions change which constraint binds: between the 2024 and 2026 data, the criterion setting the DSA target changes in six countries and the constraint setting the binding target changes in eight (Tables 1 and A7).[12] The debt sustainability safeguard no longer binds in Austria, whose four-year binding target falls from 2.58 to 1.17 percent of GDP, while Finland's rises from 5.97 to 7.44 percent of GDP as its initial debt crosses 90 percent of GDP. Moreover, on the 2026 data, the seven-year binding target at the end of the adjustment period exceeds the four-year target in 11 of the 14 countries with debt above 60 percent of GDP, even though spreading the consolidation over seven years still lowers the required annual adjustment (Table A7). In France, Belgium, Austria and Romania, the minimum EDP adjustment sets the seven-year target and accumulates 0.5-point steps over more years in the procedure. In Italy and Finland, constant annual steps under the debt sustainability safeguard keep debt above 90 percent of GDP for longer; and in Spain, Hungary, Poland, Slovenia and Slovakia, shifting the ten-year post-adjustment evaluation window from 2031–2040 to 2034–2043 exposes the projection to higher refinanced interest rates and steeper ageing costs. The average sensitivity of the rules to their inputs is thus stable over time, but which inputs matter for a given country changes whenever its binding constraint shifts.

# 8 Conclusions

How much a member state's adjustment target depends on the inputs of the Commission's DSA depends on which criterion binds. Because forecasts and model parameters shift the entire

[11] Applying the 2024 method to the 2026 data changes the binding targets by more than 0.1 percentage points only where the debt sustainability safeguard binds, raising them by 1.70 points in Finland and 0.63 points in Germany due to the higher 0.75 multiplier.

[12] For Greece, part of the change between vintages is statistical: since October 2024, deferred interest on EFSF loans is recorded in Maastricht debt as it accrues, though applying this rule to the 2024 data changes Greece's binding target by only −0.02 points over four years and +0.15 points over seven years.

baseline debt path, they affect all deterministic criteria at once and move the targets of most countries. When they are drawn jointly from past forecast errors and published estimates, the interquartile range of the binding target averages 1.07 percentage points of GDP across the 15 reference-trajectory countries with adjustment needs, running from 0.39 points below the baseline to 0.68 points above it against an average required adjustment of 3.3 points over four years. This is the range of targets that a government could defend with inputs inside past error ranges. In Greece and Portugal, this exceeds the required adjustment itself, even after demeaning historical errors to remove past trend and forecast biases.

Within the alternatives considered, the window over which the target covers projected ageing costs moves the DSA targets by amounts comparable to forecast errors. Shortening it to five years lowers the average DSA target by 0.24 points, and dropping it lowers the average target by 0.59 points. The modelling conventions governing acceptable risk move the DSA targets less: raising the probability threshold and all stress-test shocks together increases the average DSA target by 0.25 points, and larger adverse $r - g$ shocks alone raise it by 0.22 points. The specification of the stochastic analysis has almost no effect on average targets because the stochastic criterion binds only for Finland. By contrast, the permanent adverse $r - g$ stress test requires a higher SPB than the stochastic criterion in 12 of the 15 countries, because its shocks are permanent, cover a ten-year horizon, and are not country-specific. The Commission's 2026 switch to an AR bootstrap therefore leaves the targets almost unchanged.

These findings point to three priorities for the review of the framework. First, the window for covering ageing costs and the calibration of the deterministic stress tests deserve at least as much attention as the stochastic simulations, which bind for only one of the 27 member states. Second, the required decline in debt should be measured over a window after the adjustment period, or the debt sustainability safeguard in Article 7 of Regulation (EU) 2024/1263 (European Parliament and Council of the European Union, 2024) should be dropped. Because the safeguard tests debt reduction during the adjustment years, it interacts strongly with the fiscal multiplier, the output gap closure rule, near-term growth shortfalls and the front-loading of adjustment (footnote 9), and can require self-defeating consolidations when multipliers are high or when consolidation depresses potential output through hysteresis or cuts in public investment (Gottschalk et al., 2025). Lowering the multiplier to 0.6 reduces this distortion but does not remove it. Third, the degree of risk that the rules accept through conventions on stress-test sizes, probability thresholds and long-run conventions on ageing costs and productivity should be justified explicitly as a policy choice.

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# Annex

This Annex describes the DSA model as implemented in this paper and the data it uses. The code, the input workbooks and a variable-by-variable list of sources are available in the accompanying repository.

## A.1 Deterministic debt projections

The model projects the debt ratio from the reference year $T$ to 2070, following Annex A3 of the Commission's Debt Sustainability Monitor (European Commission, 2024b). The debt ratio evolves as

$$b_t = b_{t-1} \frac{1 + iir_t}{1 + g_t} \left( \alpha^n + \alpha^e \frac{e^e_t}{e^e_{t-1}} + \alpha^u \frac{e^u_t}{e^u_{t-1}} \right) - pb_t + sf_t,$$

where $b_t$ is the debt ratio, $iir_t$ the implicit interest rate on debt, $g_t$ nominal GDP growth, $pb_t$ the primary balance and $sf_t$ the stock-flow adjustment, the last two in percent of GDP. The weights $\alpha^n$, $\alpha^e$ and $\alpha^u$ are the shares of debt denominated in national currency, euro and US dollars, and $e^e_t$ and $e^u_t$ are the exchange rates against the euro and the US dollar (for euro-area members, all euro debt counts as domestic). The reference year comes from the Commission forecast, and the model projects all variables forward from the first adjustment year.

Nominal growth combines real growth and the GDP deflator. In the no-policy-change baseline, the SPB stays at its value in the last forecast year ($T + 1$), and real GDP follows the Commission forecast and then closes the output gap over three years. Changes in the SPB relative to this baseline affect real growth through the fiscal multiplier under the output gap rule described below, with a baseline multiplier of 0.75 in the 2024 method (Carnot and de Castro, 2015) and 0.6 in the 2026 method (European Commission, 2026, Box 2.5). Potential growth up to $T + 9$ is taken from the $T + 10$ projections of the Output Gaps Working Group for the 2024 data and of the Potential Output Working Group for the 2026 data, estimated with the Commission's EUCAM method, and from $T + 12$ onward from the 2024 Ageing Report (European Commission, 2024a), with linear interpolation in between. The GDP deflator follows the short-term forecast and then converges linearly to market-based inflation expectations at $T + 10$, derived from inflation swaps, and to the inflation target at $T + 30$. For Hungary, Poland and Romania, the $T + 10$ anchor adds half of the initial inflation spread over the euro area, and the $T + 30$ anchor is the national inflation target.

A change in the SPB relative to the baseline lowers the output gap in the same year by the change times the fiscal multiplier. Let $e_t$ denote this multiplier effect in year $t$ and $\tilde{y}_t$ the deviation of the output gap from its baseline. In the Commission's calculation sheets, the deviation follows a recursive rule during the adjustment,

$$\tilde{y}_t = \tfrac{2}{3} \tilde{y}_{t-1} - e_t,$$

and after the adjustment it closes linearly over three years. Because two-thirds of the whole previous deviation carry over each year, the output effects of all earlier steps keep building up for as long as the adjustment lasts. For comparison, I also evaluate an impulse-response rule in which the effect of each step fades linearly over three years regardless of later steps,

$$\tilde{y}_t = -\left(e_t + \tfrac{2}{3}e_{t-1} + \tfrac{1}{3}e_{t-2}\right).$$

With a constant step $e$ in each year of a four-year adjustment, the deviation in the fourth year reaches $-\left(1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27}\right)e \approx -2.41e$ under the recursive rule (and approaches $-3e$ over a seven-year adjustment), compared with $-2e$ under the impulse-response rule. When the persistence of the multiplier is $P$ years instead of three, the carried-over share is $(P-1)/P$ and the gap closes over $P$ years.

The primary balance is the sum of the SPB, one-off measures and a cyclical component equal to the budget semi-elasticity times the output gap. After the adjustment period, the SPB before ageing costs stays at its value in the last adjustment year. During the ten-year ageing-cost period, increases in ageing costs, net of taxes on pensions, lower the SPB one-for-one, and increases in property income raise it; both components then stay at their values in the last year of the window. Stock-flow adjustments follow the Commission's projections throughout.

The implicit interest rate is a weighted average of the short-term market rate $i_t^{ST}$ and the implicit rate on long-term debt $iir_t^{LT}$,

$$iir_t = \alpha_{t-1}\, i_t^{ST} + (1-\alpha_{t-1})\, iir_t^{LT} + a_t, \qquad iir_t^{LT} = \beta_{t-1}\, i_t^{LT} + (1-\beta_{t-1})\, iir_{t-1}^{LT},$$

where $\alpha_{t-1}$ is the share of short-term debt in total debt, $\beta_{t-1}$ the share of newly issued long-term debt in long-term debt and $i_t^{LT}$ the long-term market rate. The term $a_t$ is a correction that the Commission's sheets add for Greece to reconcile the simplified sheet calculation with the Commission's full model, and is zero for all other countries. Market rates start from benchmark rates in $T$ and $T+1$, converge linearly to ten-year-ahead forward rates at $T+10$ for the three-month and ten-year maturities, and then converge between $T+10$ and $T+30$ to their long-run anchors (equal to the inflation target plus 2 percentage points for the long-term rate and half of that for the short-term rate).

To project the shares $\alpha_t$ and $\beta_t$, the model tracks the stocks of short-term and long-term debt from annual gross financing needs, which equal interest payments plus repayments, minus the primary balance, plus the stock-flow adjustment. Short-term debt is repaid every year, while a share $s_t$ of long-term debt matures each year and is refinanced at market rates. In the Commission's sheets, $s_t$ converges from its latest value to a six-year historical average by $T+10$, based on ECB data, and reflects the repayment of official loans in Greece, Cyprus, Ireland and Portugal (in the October 2026 data, loans from the ESM and the EFSF follow their own repayment schedules and are excluded from the debt stock to which $s_t$ applies). New financing is split between short-term and long-term debt so that the short-term share stays at its average of

the last three years. In the repayment profile check of Section 7.1, long-term debt outstanding in $T$ is instead repaid according to Eurostat data on debt by residual maturity, with each maturity bucket spread evenly over its years, while new long-term debt is repaid evenly over 20 years.

The three deterministic stress tests modify the projection after the adjustment period. In the lower-SPB test, the SPB falls by 0.5 percentage points over the two years after the adjustment (or three years for a seven-year adjustment) and then stays at that level. In the adverse $r - g$ test, market interest rates are 0.5 percentage points higher and real growth is 0.5 points lower in all years after the adjustment. In the financial stress test, market interest rates rise in the first year after the adjustment by 1 percentage point, plus 0.06 points for each percentage point of debt above 90 percent of GDP.

## A.2 Stochastic debt projections

The stochastic projection follows Annex A4 of the Debt Sustainability Monitor, using the 2024 prior guidance method in the baseline (European Commission, 2024b) and the Commission's 2026 method from the Debt Sustainability Monitor 2025 in the October 2026 data (European Commission, 2026). In both methods, the projection starts in the first year after the adjustment period, covers five years and simulates 10,000 shock paths with the same random seed across runs (with a different seed, no DSA or binding target in the first 25 joint draws changes by more than 0.03 percentage points). Shocks are the first differences of quarterly series since 2000 for six variables: the exchange rates against the euro and the US dollar, the three-month and ten-year interest rates, nominal GDP growth and the primary balance, with each shock series winsorised at its 5th and 95th percentiles.

The 2024 method draws the quarterly shocks from a joint normal distribution with the covariance matrix of the historical shocks, so shocks in different quarters are independent. The 2026 method filters each shock series with its own univariate autoregressive model of order $p$, an AR($p$) model, estimated by ordinary least squares:

$$\Delta x_q = c + \sum_{j=1}^{p} \rho_j \, \Delta x_{q-j} + \varepsilon_q,$$

where $\Delta x_q$ is the shock to variable $x$ in quarter $q$ and $\varepsilon_q$ is the residual, with lag order $p = 4$ for nominal growth and $p = 1$ for the other variables in all countries. Because all series are estimated over the same quarters, each simulated path draws 20 quarters of residuals with replacement and takes the residuals of all variables from the same drawn quarter, preserving the contemporaneous correlation of shocks.[13] The drawn residuals are then passed through the estimated AR models, with the lags before the first simulated quarter set to zero.

Read literally, both Annex A4.3 of the Debt Sustainability Monitor 2025 and Bec et al. (2025)

[13] Box 2.3 of the Debt Sustainability Monitor 2025 calls this procedure a variable-block bootstrap, whereas its Annex A4.3, which I follow, specifies resampling single quarters rather than blocks of consecutive quarters.

keep the estimated constant $c$ in the simulation equation. I drop it in the baseline implementation because the Commission attributes the effects of the new method solely to the width of the debt cone (European Commission, 2026, Box 2.3), which implies mean-zero shocks keeping the fan chart centred on the deterministic projection. With the constant retained, the simulated shocks drift toward the average historical change of each series: for Finland, the constants imply that nominal growth falls by 0.45 percentage points and the primary balance by 0.32 points in each year of the projection, shifting the fan chart upward. Section 5 reports checks that keep the constant, alongside a vector autoregression with one lag, VAR(1), estimated on the six shock series and resampled in the same way (Darvas et al., 2025a).

Annual shocks are the sums of the four quarterly shocks, except for the long-term interest rate. Because a shock to this rate affects interest payments only as maturing debt is refinanced, it carries over to later years in proportion to the share of debt rolled over:

$$\varepsilon_t^{LT} = \frac{t}{m} \sum_{q=1}^{4t} \varepsilon_q^{LT},$$

where $q$ counts the quarters since the start of the stochastic projection and $m$ is the average residual maturity of debt in years (for years beyond $m$, the weight is capped at one and the sum covers the last $4m$ quarters). The maturity $m$ equals Eurostat's average residual maturity of debt where the input data include it, and otherwise equals one over the average share of long-term debt maturing each year, capped at 30 years (for Finland, replacing the resulting 26 years in the 2024 sheets with Eurostat's 7.8 years raises the standalone stochastic SPB requirement by only 0.01 percentage points). The shock to the implicit interest rate is the weighted average of the annual shocks to the short-term and long-term rates,

$$\varepsilon_t^{iir} = \alpha^{ST}\, \varepsilon_t^{ST} + (1 - \alpha^{ST})\, \varepsilon_t^{LT},$$

where $\alpha^{ST}$ is the share of short-term debt in total debt. Adding the annual shocks to the deterministic projections yields 10,000 debt paths via the identity of Section A.1. The stochastic criterion is met if debt five years after the adjustment period is at or below its level at the end of the adjustment period in at least 70 percent of the draws.

## A.3 Targets and constraints

The model computes the targets following the rules of the Commission's prior guidance calculation sheets. The DSA-based adjustment is the smallest constant annual change in the SPB for which the projection meets the DSA criteria of Section 2, found by bisection with a tolerance of 0.0001 percentage points. Whereas the Commission's sheets round this annual adjustment up to 0.01 points, the model keeps the exact value to avoid rounding differences of up to 0.04 points over four years.

For the binding target, the minimum steps of the EDP and of the deficit resilience safeguard are added on top of the constant DSA adjustment, as in the Commission's sheets. Under the EDP, the adjustment must be at least 0.5 percentage points in the SPB until 2027, or in the structural balance from 2028, in each year after a deficit above 3 percent of GDP, ending after two consecutive years with a deficit below 3 percent. Under the deficit resilience safeguard, the SPB must rise by at least 0.4 points per year (0.25 points for a seven-year adjustment) after each year with a structural deficit above 1.5 percent of GDP (implemented with a threshold of 1.55 percent in the sheets to allow for rounding).

If the resulting path does not meet the debt sustainability safeguard, the model raises the constant annual adjustment until it is met. As in the Commission's sheets, the safeguard assigns each adjustment year outside the EDP to a debt band based on the debt ratio at the start of the year: debt must decline on average by more than 1 percentage point per year over the years starting above 90 percent of GDP, and by more than 0.5 points over the years starting between 60 and 90 percent. Because a larger adjustment can lower GDP enough to push debt above 90 percent during the adjustment and trigger the stricter 1-point requirement from the following year, the safeguard is not monotonic in the adjustment. The search therefore steps up the annual adjustment by 0.01 points up to 5 percentage points and refines the threshold by bisection; runs in which no annual adjustment up to 5 percentage points meets the safeguard are counted as infeasible.

## A.4 Data

The baseline reads all macroeconomic, fiscal and institutional inputs from the Commission's 2024 prior guidance calculation sheets, supplemented by two external datasets: the historical quarterly series for the stochastic shocks (from Eurostat, supplemented with OECD short-term rates outside the euro area and ECB long-term rates for Estonia) and Eurostat data on government debt by residual maturity for the repayment profile check.

The October 2026 dataset follows the Commission's 2026 method and replaces the 2024 sheets with the latest public data, shifting the reference year to 2026 and the four-year adjustment to 2027–2030. Macroeconomic and fiscal projections up to 2027 are taken from the Commission's spring 2026 forecast in the AMECO database (retrieved through DBnomics on 4 June 2026). Real and potential GDP from 2028 to 2035 follow the spring 2026 projections of the Potential Output Working Group (estimated with the Commission's EUCAM $T + 10$ method, closing the output gap by 2030) and are interpolated linearly to the 2024 Ageing Report projections from 2038 onward. The $T + 10$ anchors for interest rates and inflation use forward rates and the euro-area five-year-in-five-year inflation swap from Bloomberg in July 2026, while market interest rates for 2026, the debt structure from the ECB and Eurostat and the historical shock series through 2026Q3 were retrieved at the end of October 2026. ESM and EFSF loan repayments come from the ESM repayment database, ageing costs from the 2024 Ageing Re-

port, stock-flow adjustments after 2027 from the country fiches of the 2025 Debt Sustainability Monitor, budget semi-elasticities from the 2024 sheets and active EDPs from Council decisions through 2025.

## A.5 Forecast errors

This section documents the construction of the forecast errors in Table 3. For the potential growth and inflation forecast errors, I use the autumn WEO vintages from 2000 to 2018 (International Monetary Fund, various years), taken from the panel compiled by Ech-charfi (2024), comparing each vintage's five-year-ahead real growth forecast (e.g., the autumn 2000 forecast for 2005) with ex-post potential growth computed from real GDP and the output gap of the October 2026 WEO. Because the WEO reports output gaps for 19 EU countries, Bulgaria, Croatia, Czechia, Hungary, Latvia, Lithuania, Poland and Romania drop out of the error estimation sample, leaving 359 observations (with France's 2019 and 2020 observations missing due to a missing 2019 output gap). The archived WEO databases report output gap projections to the fifth year only for Germany, France and Italy; for these three, the projected gap shrinks in the fifth year by 0.3 to 1.0 percentage points on average in the 2009–2015 vintages, so the real growth forecast exceeds the implied potential growth forecast by that amount before country demeaning.

For interest rates, only Germany has a public zero-coupon yield curve reaching ten years ahead since the 1980s. I use the ten-year-ahead forward rates of the Bundesbank at year ends from 1986 to 2015, with the one-year rate (the shortest maturity on the curve) standing in for the three-month rate. Because consecutive ten-year windows overlap by nine years, the 30 annual observations contain only about three independent decades.

For ageing costs, the interquartile range of the revisions across the 2015, 2018, 2021 and 2024 Ageing Reports ($-0.59$ to $+0.44$ percentage points of GDP) is about three times as wide as the range across the demographic sensitivity scenarios of the 2024 Ageing Report alone ($-0.24$ to $+0.14$ points), in part because inter-report revisions include legislated pension reforms alongside demographic updates.

**Table A1:** Standalone criterion with one group of inputs drawn at a time (percentage points of GDP)

| Drawn | P10 | P25 | Median | P75 | P90 | Mean |
|---|---|---|---|---|---|---|
| Potential growth errors | −0.95 | −0.42 | 0.00 | +0.50 | +1.08 | +0.06 |
| Interest-rate errors | −0.46 | −0.38 | −0.12 | +0.30 | +0.75 | +0.02 |
| Inflation errors | −0.23 | −0.12 | +0.02 | +0.16 | +0.29 | +0.02 |
| Ageing cost revisions | −0.44 | −0.26 | −0.05 | +0.21 | +0.50 | +0.04 |
| Model parameters | −0.03 | −0.02 | 0.00 | +0.04 | +0.07 | +0.01 |
| All jointly | −1.16 | −0.56 | +0.06 | +0.75 | +1.54 | +0.13 |

Source: Author's calculations. Note: Change in the SPB required by the standalone criterion (the baseline deterministic debt criterion) relative to the baseline. Each row draws only the stated group 500 times, as in the joint draws, and holds the other inputs at their baseline values. Interest-rate errors: short- and long-term rates. Model parameters: fiscal multiplier, its persistence and the budget semi-elasticity. P10 to P90: means of the country-level percentiles across the 15 reference-trajectory countries with adjustment needs.

**Table A2:** Baseline SPB targets and Commission prior guidance, seven-year adjustment (percent of GDP)

| | 2024 | | DSA target | | Binding target | | | |
|---|---|---|---|---|---|---|---|---|
| | Debt | SPB | Comm. | Model | Comm. | Model | DSA criterion | Binding constraint |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Greece | 153.9 | 1.73 | 2.71 | 2.48 | 2.82 | 2.62 | Stochastic | Deficit resilience |
| Italy† | 138.6 | −1.05 | 3.15 | 3.06 | 3.32 | 3.30 | Financial stress test: debt declines | EDP minimum adjustment |
| France† | 112.4 | −2.97 | 0.95 | 0.94 | 1.23 | 1.22 | Deficit below 3% | EDP minimum adjustment |
| Spain† | 105.5 | −0.85 | 2.65 | 2.64 | 2.65 | 2.64 | Adverse $r - g$ test: debt declines | DSA criteria |
| Belgium† | 103.4 | −2.03 | 1.33 | 1.22 | 1.59 | 1.56 | Adverse $r - g$ test: debt declines | EDP minimum adjustment |
| Portugal | 95.6 | 2.20 | 2.48 | 2.24 | 2.48 | 2.24 | Adverse $r - g$ test: debt declines | DSA criteria |
| Finland† | 80.5 | −0.53 | 1.57 | 1.46 | 4.79 | 4.78 | Adverse $r - g$ test: debt declines | Debt safeguard |
| Austria† | 79.5 | −1.34 | 0.90 | 0.59 | 1.88 | 1.87 | Adverse $r - g$ test: debt declines | Debt safeguard |
| Hungary | 74.3 | 0.03 | 2.97 | 2.89 | 3.31 | 3.26 | Adverse $r - g$ test: debt below 60% | EDP minimum adjustment |
| Cyprus | 70.6 | 3.48 | 3.48 | −0.04 | 3.48 | 3.48 | Deficit below 3% | Adjustment floor |
| Slovenia | 68.1 | −1.24 | 0.58 | 0.45 | 0.58 | 0.49 | Lower-SPB test: debt below 60% | Deficit resilience |
| Germany† | 62.5 | −1.02 | 0.80 | 0.76 | 2.20 | 2.16 | Lower-SPB test: debt declines | Debt safeguard |
| Croatia | 59.5 | −1.96 | – | −0.31 | −0.21 | −0.21 | Baseline: debt below 60% | Deficit resilience |
| Slovakia | 58.5 | −4.26 | 1.34 | 1.20 | 1.34 | 1.20 | Lower-SPB test: debt below 60% | DSA criteria |
| Poland | 53.7 | −2.61 | 0.75 | 0.77 | 1.19 | 1.18 | Deficit below 3% | EDP minimum adjustment |
| Malta | 52.0 | −2.90 | −0.94 | −0.96 | −0.03 | −0.04 | Deficit below 3% | EDP minimum adjustment |
| Romania† | 50.9 | −4.41 | 0.77 | 0.79 | 0.78 | 0.79 | Deficit below 3% | DSA criteria |
| Netherlands | 47.1 | −0.64 | – | −0.03 | 0.06 | −0.03 | Deficit below 3% | DSA criteria |
| Czechia | 45.2 | −0.09 | – | 0.66 | 0.75 | 0.66 | Deficit below 3% | DSA criteria |
| Latvia | 44.5 | −1.36 | – | −1.13 | 0.18 | 0.17 | Baseline: debt below 60% | Deficit resilience |
| Ireland | 42.5 | 2.53 | – | −0.64 | – | −0.64 | Deficit below 3% | DSA criteria |
| Lithuania | 38.3 | −0.37 | – | 0.09 | 0.19 | 0.09 | Deficit below 3% | DSA criteria |
| Sweden | 32.0 | 0.69 | – | −1.87 | −0.85 | −0.64 | Deficit below 3% | Deficit resilience |
| Luxembourg | 27.1 | 0.10 | – | −0.32 | – | −0.32 | Deficit below 3% | DSA criteria |
| Denmark | 26.5 | 2.91 | – | −1.78 | −1.15 | −1.78 | Deficit below 3% | DSA criteria |
| Bulgaria | 24.5 | −2.09 | – | −1.66 | −0.53 | −0.53 | Deficit below 3% | Deficit resilience |
| Estonia | 21.4 | −0.29 | −0.29 | −1.72 | −0.29 | −0.29 | Deficit below 3% | Adjustment floor |

Source: Author's calculations based on the European Commission's prior guidance calculation sheets of 2024.
Note: SPB at the end of a seven-year adjustment period (2031); see Table 1. Column (3) is the SPB in 2024 plus seven times the DSA-based annual adjustment. The largest difference in the DSA target is 0.31 percentage points (Austria), and in the binding target 0.63 points (Denmark, where the Commission's seven-year technical information equals the 2024 SPB minus about 4 percentage points as over four years, whereas the model applies the floor of −1 point per year, or −7 points over seven years, so that the deficit criterion sets the target at −1.78).
†Medium-term plan with a seven-year adjustment period.

**Table A3a:** Change in the SPB relative to the baseline: forecasts and model parameters, seven-year adjustment (percentage points of GDP)

| | | Standalone criterion | | DSA target | | Binding target | |
|---|---|---|---|---|---|---|---|
| Input | Value | Mean | Largest | Mean | Largest | Mean | Largest |
| *1. Forecasts* | | | | | | | |
| Real short-term rate, $T + 10$ | −1 pp. | −0.05 | −0.15 ITA | −0.05 | −0.13 ITA | −0.04 | −0.14 ITA |
| | +1 pp. | +0.05 | +0.16 ITA | +0.05 | +0.13 ITA | +0.04 | +0.14 ITA |
| Real long-term rate, $T + 10$ | −1 pp. | −0.32 | −0.64 GRC | −0.26 | −0.47 BEL | −0.16 | −0.44 ESP |
| | +1 pp. | +0.33 | +0.59 GRC | +0.29 | +0.55 ITA | +0.21 | +0.44 GRC |
| Inflation, $T + 10$ | −0.5 pp. | +0.11 | +0.30 ITA | +0.09 | +0.34 GRC | +0.06 | +0.29 GRC |
| | +0.5 pp. | −0.10 | −0.28 ITA | −0.02 | −0.22 FIN | −0.02 | −0.20 FIN |
| Potential growth, $T + 10$ | −0.75 pp. | +0.45 | +1.00 ITA | +0.41 | +0.76 GRC | +0.31 | +0.65 GRC |
| | +0.75 pp. | −0.43 | −0.82 ITA | −0.27 | −0.52 GRC | −0.18 | −0.40 FIN |
| Ageing costs | −0.5 pp. | −0.19 | −0.36 BEL | −0.15 | −0.34 BEL | −0.06 | −0.33 ESP |
| | +0.5 pp. | +0.20 | +0.37 AUT | +0.21 | +0.34 AUT | +0.12 | +0.33 PRT |
| Real short-term rate, $T + 30$ | −0.5 pp. | −0.01 | −0.02 PRT | 0.00 | −0.02 PRT | 0.00 | −0.02 PRT |
| | +0.5 pp. | +0.01 | +0.02 PRT | 0.00 | +0.02 PRT | 0.00 | +0.02 PRT |
| Real long-term rate, $T + 30$ | −0.5 pp. | −0.02 | −0.07 GRC | −0.01 | −0.05 AUT | −0.01 | −0.04 ESP |
| | +0.5 pp. | +0.02 | +0.07 GRC | +0.02 | +0.05 AUT | +0.01 | +0.04 ESP |
| Inflation, $T + 30$ | −0.25 pp. | +0.02 | +0.06 GRC | +0.01 | +0.05 BEL | +0.01 | +0.04 ESP |
| | +0.25 pp. | −0.02 | −0.06 GRC | −0.01 | −0.05 BEL | −0.01 | −0.04 ESP |
| Ageing costs, previous vintage | 2021 Ageing Report | +0.04 | −1.04 PRT | +0.16 | +1.29 SVN | +0.13 | +1.25 SVN |
| Ageing costs, 2024 AR scenario | higher migration | −0.14 | −0.48 ESP | −0.11 | −0.44 ESP | −0.05 | −0.44 ESP |
| | lower fertility | −0.17 | −0.43 PRT | −0.14 | −0.40 PRT | −0.05 | −0.23 ESP |
| *2. Model parameters* | | | | | | | |
| Fiscal multiplier | 0.5 | −0.02 | −0.09 SVK | −0.06 | −0.30 FRA | −0.13 | −0.66 FIN |
| | 0.6 | −0.01 | −0.05 SVK | −0.04 | −0.19 FRA | −0.09 | −0.41 FIN |
| | 1 | +0.02 | +0.09 SVK | +0.08 | +0.37 FRA | +0.18 | +0.88 FIN |
| | 1.5 | +0.08 | +0.28 SVK | +0.33 | +1.38 FRA | +0.84 | +4.00 FIN |
| Multiplier persistence | 2 years | −0.02 | +0.13 ITA | −0.06 | −0.22 FRA | −0.15 | −0.80 FIN |
| | 4 years | +0.05 | +0.17 FRA | +0.07 | +0.22 FRA | +0.14 | +0.71 FIN |
| | 5 years | +0.10 | +0.33 FRA | +0.14 | +0.40 FRA | +0.25 | +1.31 FIN |
| Impulse-response output gap rule | 2 years | −0.03 | −0.14 FIN | −0.08 | −0.23 FRA | −0.20 | −1.07 FIN |
| | 3 years | −0.03 | −0.11 FRA | −0.06 | −0.25 FRA | −0.14 | −0.78 FIN |
| | 4 years | −0.01 | −0.06 FRA | −0.03 | −0.11 ROU | −0.07 | −0.48 FIN |
| | 5 years | +0.01 | +0.09 ITA | +0.01 | +0.09 FRA | −0.01 | −0.17 FIN |
| Budget semi-elasticity | ×0.9 | −0.01 | −0.04 FRA | −0.02 | −0.09 FRA | −0.06 | −0.28 FIN |
| | ×1.1 | +0.01 | +0.04 FRA | +0.03 | +0.10 FRA | +0.06 | +0.30 FIN |

Source: Author's calculations. Note: See Table 4. With a seven-year adjustment, the debt sustainability safeguard can be met in all runs.

**Table A3b:** Change in the SPB relative to the baseline: estimation choices, calibration conventions and data, seven-year adjustment (percentage points of GDP)

| | | Standalone criterion | | DSA target | | Binding target | |
|---|---|---|---|---|---|---|---|
| Input | Value | Mean | Largest | Mean | Largest | Mean | Largest |
| *3. Estimation choices* | | | | | | | |
| Start of shock sample | 1990 | −0.01 | −0.03 FRA | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | 2010 | +0.01 | +0.37 SVN | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| Shock frequency | annual | −0.15 | −0.57 MLT | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| Shock model | VAR bootstrap | −0.16 | −0.52 MLT | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | AR bootstrap | −0.26 | −0.60 MLT | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | VAR bootstrap, intercept | +0.05 | −0.70 ROU | +0.05 | +0.57 GRC | +0.03 | +0.49 GRC |
| | AR bootstrap, intercept | −0.03 | −0.66 MLT | +0.03 | +0.24 GRC | +0.01 | +0.20 GRC |
| | AR bootstrap, not winsorised | +0.13 | +0.65 ITA | +0.03 | +0.47 GRC | +0.03 | +0.40 GRC |
| Winsorised shocks | P1 and P99 | +0.52 | +1.13 ESP | +0.12 | +0.48 AUT | +0.05 | +0.26 GRC |
| | off | +0.83 | +1.66 ESP | +0.29 | +1.11 GRC | +0.16 | +0.97 GRC |
| *4. Calibration conventions* | | | | | | | |
| Start of stochastic projection | first adjustment year | +0.45 | +1.88 GRC | +0.17 | +1.88 GRC | +0.13 | +1.74 GRC |
| Horizon of stochastic projection | 10 years | −0.02 | −0.84 FIN | 0.00 | +0.04 GRC | 0.00 | +0.04 GRC |
| Threshold, start, horizon | 60%, after adj., 5 years | −0.50 | −0.89 GRC | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | 80%, after adj., 5 years | +0.57 | +0.95 GRC | +0.13 | +0.95 GRC | +0.08 | +0.81 GRC |
| | 60%, after adj., 10 years | −0.41 | −1.12 FIN | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | 80%, after adj., 10 years | +0.43 | +0.87 GRC | +0.08 | +0.87 GRC | +0.06 | +0.75 GRC |
| | 60%, first adj. year, 5 years | −0.32 | −0.73 MLT | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | 80%, first adj. year, 5 years | +1.36 | +4.24 GRC | +0.57 | +4.06 GRC | +0.41 | +3.92 GRC |
| | 60%, first adj. year, 10 years | −0.26 | −1.06 FIN | 0.00 | −0.02 GRC | 0.00 | −0.02 GRC |
| | 70%, first adj. year, 10 years | +0.31 | +1.45 GRC | +0.10 | +1.45 GRC | +0.09 | +1.31 GRC |
| | 80%, first adj. year, 10 years | +0.97 | +3.13 GRC | +0.35 | +3.13 GRC | +0.26 | +2.98 GRC |
| Adverse $r - g$ shock | 0.25 pp. | −0.27 | −0.50 ITA | −0.05 | −0.25 FIN | −0.02 | −0.18 ESP |
| | 0.75 pp. | +0.27 | +0.49 ITA | +0.20 | +0.42 GRC | +0.10 | +0.36 GRC |
| Lower-SPB shock | 0.25 pp. | −0.20 | −0.26 DEU | −0.01 | −0.05 SVK | 0.00 | −0.05 SVK |
| | 0.75 pp. | +0.20 | +0.26 DEU | +0.08 | +0.26 DEU | +0.05 | +0.23 PRT |
| Financial stress shock | 0.5 pp. | −0.03 | −0.14 ITA | 0.00 | −0.03 ITA | 0.00 | −0.01 ITA |
| | 1.5 pp. | +0.03 | +0.14 ITA | +0.01 | +0.14 ITA | 0.00 | +0.04 ITA |
| Ageing cost period | 0 years | −0.56 | −1.99 ESP | −0.48 | −1.90 ESP | −0.20 | −1.41 ESP |
| | 5 years | −0.21 | −0.87 ESP | −0.16 | −0.82 ESP | −0.09 | −0.82 ESP |
| TFP growth after $T + 10$ | held at $T + 10$ value | +0.25 | +0.72 GRC | +0.21 | +0.63 ESP | +0.14 | +0.63 ESP |
| All settings | lenient | −0.21 | −0.87 ESP | −0.41 | −1.16 ESP | −0.16 | −1.16 ESP |
| | strict | +0.25 | +0.72 GRC | +0.46 | +1.03 GRC | +0.31 | +0.92 ESP |
| Threshold and stress tests | lenient | – | – | −0.16 | −0.33 ESP | −0.05 | −0.33 ESP |
| | strict | – | – | +0.26 | +0.95 GRC | +0.15 | +0.81 GRC |
| *Data* | | | | | | | |
| Repayment profile | Eurostat profile | −0.02 | −0.49 GRC | −0.01 | −0.33 GRC | +0.03 | +0.45 FIN |

Source: Author's calculations. Note: See Table 6. With a seven-year adjustment, the debt sustainability safeguard can be met in all runs.

**Table A4:** Change in the SPB relative to the baseline: other combinations of threshold, start and horizon of the stochastic criterion, four-year adjustment (percentage points of GDP)

| | | Standalone criterion | | DSA target | | Binding target | |
|---|---|---|---|---|---|---|---|
| Input | Value | Mean | Largest | Mean | Largest | Mean | Largest |
| *4. Calibration conventions* | | | | | | | |
| Threshold, start, horizon | 60%, after adj., 10 years | −0.28 | −0.96 FIN | −0.01 | −0.08 FIN | 0.00 | 0.00 |
| | 80%, after adj., 10 years | +0.56 | +1.46 GRC | +0.04 | +0.55 GRC | +0.03 | +0.41 GRC |
| | 60%, first adj. year, 5 years | −0.44 | −0.79 MLT | −0.01 | −0.08 FIN | 0.00 | 0.00 |
| | 60%, first adj. year, 10 years | −0.20 | −0.92 FIN | −0.01 | −0.08 FIN | 0.00 | 0.00 |
| | 70%, first adj. year, 10 years | +0.30 | +1.51 GRC | +0.03 | +0.59 GRC | +0.03 | +0.45 GRC |

Source: Author's calculations. Note: See Table 6, which reports the other four combinations.

**Table A5:** Effect of selected calibration conventions by country, DSA target, four-year adjustment (percentage points of GDP)

| | Thr. 80% (1) | Early start (2) | Early, 80% (3) | $r-g$ 0.75 pp. (4) | Low SPB 0.75 pp. (5) | Ageing 5 yrs (6) | Ageing 0 yrs (7) | TFP held (8) | Lenient, margins (9) | Strict, margins (10) | Lenient (11) | Strict (12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Greece | +0.08 | 0.00 | +1.55 | +0.44 | 0.00 | −0.41 | −0.80 | +0.44 | −0.28 | +0.44 | −0.63 | +0.86 |
| Italy | 0.00 | 0.00 | +0.21 | +0.50 | 0.00 | 0.00 | −0.57 | +0.22 | −0.19 | +0.50 | −0.38 | +0.77 |
| France | 0.00 | 0.00 | +0.28 | +0.29 | 0.00 | 0.00 | 0.00 | +0.30 | 0.00 | +0.29 | 0.00 | +0.73 |
| Spain | 0.00 | 0.00 | 0.00 | +0.34 | +0.05 | −0.89 | −1.83 | +0.44 | −0.34 | +0.34 | −1.26 | +0.76 |
| Belgium | 0.00 | 0.00 | 0.00 | +0.37 | 0.00 | −0.49 | −0.59 | +0.35 | −0.27 | +0.37 | −0.59 | +0.70 |
| Portugal | 0.00 | 0.00 | +0.21 | +0.27 | +0.19 | −0.77 | −1.43 | +0.25 | −0.27 | +0.27 | −1.06 | +0.51 |
| Finland | +0.43 | +0.06 | +0.55 | +0.17 | +0.12 | 0.00 | −0.19 | 0.00 | −0.33 | +0.43 | −0.33 | +0.43 |
| Austria | +0.13 | 0.00 | +0.33 | +0.31 | +0.11 | −0.22 | −0.63 | +0.17 | −0.32 | +0.31 | −0.55 | +0.48 |
| Hungary | +0.06 | 0.00 | +0.06 | +0.23 | +0.24 | −0.45 | −0.60 | 0.00 | −0.24 | +0.24 | −0.93 | +0.24 |
| Cyprus | 0.00 | 0.00 | 0.00 | +0.07 | +0.07 | −0.31 | −0.80 | +0.01 | 0.00 | +0.07 | −0.49 | +0.11 |
| Slovenia | 0.00 | 0.00 | 0.00 | +0.11 | +0.19 | −0.21 | −0.79 | 0.00 | −0.14 | +0.19 | −0.40 | +0.19 |
| Germany | 0.00 | 0.00 | 0.00 | +0.18 | +0.19 | −0.07 | −0.69 | +0.02 | −0.25 | +0.19 | −0.32 | +0.21 |
| Croatia | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.05 | +0.13 | +0.04 | 0.00 | 0.00 | +0.05 | +0.04 |
| Slovakia | 0.00 | 0.00 | 0.00 | +0.08 | +0.19 | −0.15 | −0.65 | 0.00 | −0.19 | +0.19 | −0.34 | +0.19 |
| Poland | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Malta | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Romania | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Netherlands | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | −0.65 | −1.54 | +0.01 | 0.00 | 0.00 | −0.65 | +0.01 |
| Czechia | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | −0.67 | −1.23 | +0.01 | 0.00 | 0.00 | −0.67 | +0.01 |
| Latvia | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.20 | +0.29 | 0.00 | 0.00 | 0.00 | +0.20 | 0.00 |
| Ireland | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | −0.46 | −0.99 | 0.00 | 0.00 | 0.00 | −0.46 | 0.00 |
| Lithuania | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | −0.54 | −1.39 | 0.00 | 0.00 | 0.00 | −0.54 | 0.00 |
| Sweden | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.05 | −0.16 | 0.00 | 0.00 | 0.00 | +0.05 | 0.00 |
| Luxembourg | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | −0.71 | −1.55 | +0.01 | 0.00 | 0.00 | −0.71 | +0.01 |
| Denmark | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | −0.16 | −0.63 | 0.00 | 0.00 | 0.00 | −0.16 | 0.00 |
| Bulgaria | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.26 | +0.69 | 0.00 | 0.00 | 0.00 | +0.26 | 0.00 |
| Estonia | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.07 | +0.26 | 0.00 | 0.00 | 0.00 | +0.07 | 0.00 |
| Mean, reference trajectory (15) | +0.05 | 0.00 | +0.21 | +0.22 | +0.09 | −0.24 | −0.59 | +0.15 | −0.19 | +0.25 | −0.45 | +0.41 |
| Mean, all with adjustment needs (21) | +0.03 | 0.00 | +0.15 | +0.16 | +0.06 | −0.24 | −0.56 | +0.11 | −0.13 | +0.18 | −0.39 | +0.29 |

Source: Author's calculations. Note: Change in the DSA target relative to the baseline. (1) Probability threshold of 80 percent. (2) Stochastic projection starting in the first adjustment year. (3) Both. (4) Adverse $r-g$ shocks of 0.75 percentage points. (5) Lower-SPB shock of 0.75 percentage points. (6) and (7) Ageing cost period of five and zero years. (8) TFP growth held at its $T+10$ value. (9) and (10) Lenient and strict scenarios for the probability threshold and the stress tests only. (11) and (12) Lenient and strict scenarios for all settings; see Table 6. The last two rows give the means across the 15 reference-trajectory countries with adjustment needs and across all 21 countries with adjustment needs. Countries are ordered by debt in 2024.

**Table A6:** Mean change in the SPB by individual check with the 2024 and the October 2026 data, four-year adjustment (percentage points of GDP)

| | | | Standalone criterion | | DSA target | | Binding target | |
|---|---|---|---|---|---|---|---|---|
| Setting | Alternative | Data | 2024 | 2026 | 2024 | 2026 | 2024 | 2026 |
| *1. Forecasts* | | | | | | | | |
| Real short-term rate, $T + 10$ | −1 pp. | | −0.05 | −0.05 | −0.04 | −0.05 | −0.04 | −0.04 |
| | +1 pp. | | +0.05 | +0.05 | +0.04 | +0.05 | +0.04 | +0.04 |
| Real long-term rate, $T + 10$ | −1 pp. | | −0.26 | −0.28 | −0.23 | −0.27 | −0.17 | −0.23 |
| | +1 pp. | | +0.29 | +0.27 | +0.22 | +0.27 | +0.18 | +0.19 |
| Inflation, $T + 10$ | −0.5 pp. | | +0.13 | +0.14 | +0.10 | +0.14 | +0.10 | +0.12 |
| | +0.5 pp. | | −0.11 | −0.13 | −0.06 | −0.11 | −0.06 | −0.10 |
| Potential growth, $T + 10$ | −0.75 pp. | | +0.46 | +0.46 | +0.38 | +0.44 | +0.33 | +0.38 |
| | +0.75 pp. | | −0.41 | −0.47 | −0.28 | −0.38 | −0.21 | −0.33 |
| Ageing costs | −0.5 pp. | | −0.22 | −0.23 | −0.19 | −0.23 | −0.13 | −0.19 |
| | +0.5 pp. | | +0.25 | +0.23 | +0.18 | +0.24 | +0.14 | +0.16 |
| Real short-term rate, $T + 30$ | −0.5 pp. | | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | +0.5 pp. | | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Real long-term rate, $T + 30$ | −0.5 pp. | | −0.01 | −0.01 | −0.01 | −0.01 | −0.01 | −0.01 |
| | +0.5 pp. | | +0.01 | +0.01 | +0.01 | +0.01 | +0.01 | +0.01 |
| Inflation, $T + 30$ | −0.25 pp. | | +0.01 | +0.01 | +0.01 | +0.01 | +0.01 | +0.01 |
| | +0.25 pp. | | −0.01 | −0.01 | −0.01 | −0.01 | −0.01 | −0.01 |
| Ageing costs, previous vintage | 2021 Ageing Report | | +0.05 | +0.02 | +0.05 | +0.07 | +0.07 | +0.11 |
| Ageing costs, 2024 AR scenario | higher migration | | −0.15 | −0.16 | −0.13 | −0.16 | −0.09 | −0.12 |
| | lower fertility | | −0.16 | −0.21 | −0.15 | −0.21 | −0.11 | −0.17 |
| *2. Model parameters* | | | | | | | | |
| Fiscal multiplier | 0.5 | | −0.02 | −0.01 | −0.08 | −0.01 | −0.26 | −0.09 |
| | 0.6 | | −0.01 | – | −0.05 | – | −0.17 | – |
| | 1 | | +0.02 | +0.03 | +0.12 | +0.13 | +0.45 | +0.81 |
| | 1.5 | | +0.06 | +0.07 | +0.63 | +0.62 | +1.35 | +1.95 |
| Multiplier persistence | 2 years | | −0.02 | −0.03 | −0.07 | −0.04 | −0.22 | −0.15 |
| | 4 years | | +0.05 | +0.03 | +0.07 | +0.04 | +0.17 | +0.11 |
| | 5 years | | +0.10 | +0.05 | +0.12 | +0.08 | +0.30 | +0.20 |
| Impulse-response output gap rule | 2 years | | −0.02 | −0.04 | −0.10 | −0.05 | −0.29 | −0.21 |
| | 3 years | | −0.02 | −0.03 | −0.06 | −0.03 | −0.17 | −0.12 |
| | 4 years | | −0.01 | −0.02 | 0.00 | −0.02 | −0.02 | −0.03 |
| | 5 years | | +0.02 | 0.00 | +0.05 | +0.01 | +0.09 | +0.03 |
| Budget semi-elasticity | ×0.9 | | −0.01 | −0.01 | −0.03 | −0.01 | −0.10 | −0.06 |
| | ×1.1 | | +0.01 | +0.01 | +0.03 | +0.01 | +0.11 | +0.06 |
| *3. Estimation choices* | | | | | | | | |
| Start of shock sample | 1990 | | −0.01 | +0.01 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2010 | | +0.02 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Shock frequency | annual | | −0.15 | −0.05 | +0.01 | 0.00 | 0.00 | 0.00 |
| Shock model | VAR bootstrap | | −0.16 | +0.10 | 0.00 | 0.00 | 0.00 | 0.00 |
| | AR bootstrap | | −0.26 | – | −0.01 | – | 0.00 | – |
| | VAR bootstrap, intercept | | +0.05 | +0.31 | +0.03 | +0.04 | 0.00 | 0.00 |
| | AR bootstrap, intercept | | −0.02 | +0.23 | +0.03 | +0.02 | 0.00 | 0.00 |
| | AR bootstrap, not winsorised | | +0.13 | – | +0.01 | – | 0.00 | – |
| Winsorised shocks | P1 and P99 | | +0.53 | +0.27 | +0.05 | 0.00 | +0.02 | 0.00 |
| | off | | +0.84 | +0.41 | +0.15 | +0.01 | +0.07 | 0.00 |
| *4. Calibration conventions* | | | | | | | | |
| Start of stochastic projection | first adjustment year | | +0.19 | +0.38 | 0.00 | +0.01 | 0.00 | 0.00 |
| Horizon of stochastic projection | 10 years | | +0.10 | +0.29 | −0.01 | 0.00 | 0.00 | 0.00 |
| Probability threshold | 60% | | −0.50 | −0.40 | −0.01 | 0.00 | 0.00 | 0.00 |
| | 80% | | +0.57 | +0.47 | +0.05 | +0.03 | +0.01 | 0.00 |
| Adverse $r - g$ shock | 0.25 pp. | | −0.29 | −0.28 | −0.08 | −0.11 | −0.06 | −0.08 |
| | 0.75 pp. | | +0.28 | +0.28 | +0.22 | +0.25 | +0.17 | +0.19 |
| Financial stress shock | 0.5 pp. | | −0.03 | −0.02 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 1.5 pp. | | +0.03 | +0.02 | 0.00 | 0.00 | 0.00 | 0.00 |
| Lower-SPB shock | 0.25 pp. | | −0.23 | −0.23 | −0.01 | −0.01 | −0.01 | −0.01 |
| | 0.75 pp. | | +0.23 | +0.23 | +0.09 | +0.12 | +0.06 | +0.10 |
| Ageing cost period | 0 years | | −0.65 | −0.69 | −0.59 | −0.67 | −0.34 | −0.45 |
| | 5 years | | −0.27 | −0.24 | −0.24 | −0.24 | −0.18 | −0.20 |
| TFP growth after $T + 10$ | held at $T + 10$ value | | +0.15 | +0.13 | +0.15 | +0.13 | +0.13 | +0.10 |
| Scenario | lenient | | −0.27 | −0.24 | −0.45 | −0.49 | −0.27 | −0.37 |
| | strict | | +0.15 | +0.13 | +0.41 | +0.39 | +0.32 | +0.32 |
| | lenient, margins only | | 0.00 | 0.00 | −0.19 | −0.24 | −0.12 | −0.19 |
| | strict, margins only | | 0.00 | 0.00 | +0.25 | +0.27 | +0.18 | +0.21 |
| *Data* | | | | | | | | |
| Repayment profile | Eurostat profile | | −0.01 | +0.05 | −0.02 | +0.06 | +0.01 | +0.05 |

Source: Author's calculations. Note: Mean change across the 15 reference-trajectory countries with adjustment needs in the 2024 data, relative to the baseline of each data vintage. 2024: Commission prior guidance data, adjustment from 2025 to 2028, with the Commission's 2024 method (fiscal multiplier of 0.75, normal shocks). 2026: October 2026 data, adjustment from 2027 to 2030, with the Commission's 2026 method (fiscal multiplier of 0.6, AR bootstrap). Checks of the 2026 method's own multiplier and shock model are left blank for the 2026 data. With a multiplier of 1.5, the debt sustainability safeguard cannot be met in some countries, which are left out of the means of the binding target.

**Table A7:** SPB targets with the October 2026 data (percent of GDP and percentage points)

| | 2026 | | Four-year adjustment | | | | | | Seven-year |
|---|---|---|---|---|---|---|---|---|---|
| | Debt | SPB | DSA target | DSA criterion | Binding target | Binding constraint | Annual adj. | Change vs. 2024 data | Binding target |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Greece | 140.7 | 2.54 | 1.98 | Adverse $r-g$ test: debt declines | 2.54 | Adjustment floor | 0.00 | −0.02 | 2.54 |
| Italy† | 138.5 | 0.97 | 4.00 | Adverse $r-g$ test: debt declines | 4.00 | DSA criteria | 0.76 | +0.70 | 4.15 |
| France† | 118.1 | −1.97 | 1.80 | Adverse $r-g$ test: debt declines | 1.80 | DSA criteria | 0.94 | +0.92 | 2.65 |
| Belgium† | 110.5 | −1.96 | 1.90 | Adverse $r-g$ test: debt declines | 1.90 | DSA criteria | 0.96 | +0.72 | 2.34 |
| Spain† | 99.6 | −0.40 | 2.84 | Adverse $r-g$ test: debt declines | 2.84 | DSA criteria | 0.81 | +0.27 | 2.97 |
| Finland† | 91.2 | −1.31 | 1.92 | Adverse $r-g$ test: debt declines | 7.44 | Debt safeguard | 2.19 | +1.47 | 10.42 |
| Portugal | 87.6 | 2.20 | 2.67 | Adverse $r-g$ test: debt declines | 2.67 | DSA criteria | 0.12 | +0.18 | 2.45 |
| Austria† | 83.4 | −1.72 | 1.17 | Adverse $r-g$ test: debt declines | 1.17 | DSA criteria | 0.72 | −1.41 | 1.66 |
| Hungary | 75.1 | −2.01 | 3.21 | Adverse $r-g$ test: debt declines | 3.21 | DSA criteria | 1.31 | +0.56 | 3.61 |
| Germany† | 65.8 | −1.77 | 1.22 | Lower-SPB test: debt declines | 2.55 | Debt safeguard | 1.08 | −0.22 | 2.49 |
| Slovenia | 64.9 | −2.40 | 0.97 | Lower-SPB test: debt below 60% | 0.97 | DSA criteria | 0.84 | +0.50 | 1.14 |
| Poland | 64.5 | −3.77 | 1.76 | Adverse $r-g$ test: debt declines | 1.76 | DSA criteria | 1.38 | +1.02 | 2.29 |
| Slovakia | 63.7 | −2.63 | 1.74 | Lower-SPB test: debt below 60% | 1.74 | DSA criteria | 1.09 | +0.85 | 1.87 |
| Romania† | 61.6 | −2.39 | 1.59 | Lower-SPB test: debt below 60% | 1.59 | DSA criteria | 1.00 | +1.05 | 2.29 |
| Croatia | 55.9 | −1.74 | −0.74 | Baseline: debt below 60% | −0.14 | Deficit resilience | 0.40 | +0.22 | 0.01 |
| Cyprus | 50.4 | 2.52 | 0.09 | Deficit below 3% | 0.09 | DSA criteria | −0.61 | −3.39 | −0.41 |
| Latvia | 48.8 | −2.08 | 0.00 | Lower-SPB test: debt below 60% | 0.00 | DSA criteria | 0.52 | +0.14 | 0.25 |
| Netherlands | 46.9 | −0.45 | 0.01 | Deficit below 3% | 0.01 | DSA criteria | 0.11 | −0.06 | −0.30 |
| Malta | 46.2 | −0.29 | −0.99 | Deficit below 3% | 0.16 | Deficit resilience | 0.11 | +1.03 | 0.00 |
| Czechia | 45.8 | −1.33 | 1.01 | Deficit below 3% | 1.01 | DSA criteria | 0.59 | +0.63 | 1.40 |
| Lithuania | 44.6 | −1.05 | 0.49 | Deficit below 3% | 0.53 | Deficit resilience | 0.40 | +0.28 | 0.55 |
| Sweden | 36.6 | −1.35 | −1.54 | Deficit below 3% | −0.64 | Deficit resilience | 0.18 | +0.11 | −0.48 |
| Ireland | 32.4 | 2.09 | −0.51 | Deficit below 3% | −0.51 | DSA criteria | −0.65 | +0.15 | −0.46 |
| Bulgaria | 32.3 | −2.93 | −1.20 | Deficit below 3% | −1.20 | DSA criteria | 0.43 | −0.31 | −0.79 |
| Luxembourg | 29.2 | 0.75 | 0.16 | Deficit below 3% | 0.16 | DSA criteria | −0.15 | +0.39 | 0.03 |
| Denmark | 27.0 | 2.75 | −1.28 | Deficit below 3% | −1.25 | Adjustment floor | −1.00 | −0.17 | −0.84 |
| Estonia | 26.9 | −3.20 | −1.22 | Deficit below 3% | −1.22 | DSA criteria | 0.49 | −0.94 | −0.88 |

Source: Author's calculations based on the October 2026 input data and the Commission's 2026 method (fiscal multiplier of 0.6, AR bootstrap). Note: Columns (1) and (2) are debt and the SPB in 2026. Columns (3) to (8) refer to a four-year adjustment period ending in 2030: the DSA target and its setting criterion, the binding target and its setting constraint, the annual adjustment required by the binding target, and the change in the binding target relative to the 2024 baseline (ending in 2028). Column (9) is the binding target at the end of a seven-year adjustment period (2033); Section 7.2 discusses why the seven-year target exceeds the four-year target in 11 of the 14 countries with debt above 60 percent of GDP. For Finland, replacing the constant annual step (10.42 percent target) with linearly declining steps lowers the seven-year target to 7.37 percent, and applying the Article 7(2) reading of Darvas et al. (2024) lowers it to 6.57 percent (footnote 9). Countries are ordered by debt in 2026. †Medium-term plan with a seven-year adjustment period.

**Table A8:** Fiscal multiplier checks under the Commission's recursive output gap rule and the impulse-response rule, binding target, four-year adjustment (percent of GDP and percentage points)

| | Austria | | Finland | | Germany | |
|---|---|---|---|---|---|---|
| | Commission | Impulse-response | Commission | Impulse-response | Commission | Impulse-response |
| Baseline target | 2.58 | 2.21 | 5.97 | 4.80 | 2.77 | 2.46 |
| Fiscal multiplier: 0.5 | −0.84 | −0.65 | −1.42 | −1.00 | −0.65 | −0.52 |
| Fiscal multiplier: 0.6 | −0.55 | −0.42 | −0.94 | −0.65 | −0.42 | −0.33 |
| Fiscal multiplier: 1 | +1.48 | +1.04 | +2.61 | +1.64 | +1.00 | +0.75 |
| Fiscal multiplier: 1.5 | inf. | +8.56 | inf. | inf. | +6.99 | +4.07 |
| Multiplier persistence: 2 years | −0.65 | −0.42 | −1.35 | −0.65 | −0.58 | −0.33 |
| Multiplier persistence: 4 years | +0.48 | +0.44 | +1.05 | +0.69 | +0.40 | +0.32 |
| Multiplier persistence: 5 years | +0.84 | +0.77 | +1.88 | +1.20 | +0.69 | +0.55 |

Source: Author's calculations. Note: The first row is the binding target of the baseline; the other rows report the change in the binding target relative to the baseline for the three countries where the debt sustainability safeguard binds in the baseline. inf.: the debt sustainability safeguard cannot be met with an annual adjustment up to 5 percentage points. Commission: recursive output gap rule of the Commission's prior guidance calculation sheets. Impulse-response: effect of each SPB change on the output gap fading linearly over the persistence horizon (Annex A.1).

**Table A9:** Binding target with real growth in the first adjustment year shifted (percent of GDP)

| | Four-year adjustment | | | Seven-year adjustment | | |
|---|---|---|---|---|---|---|
| | Baseline | −1.25 pp. | +1.25 pp. | Baseline | −1.25 pp. | +1.25 pp. |
| Greece | 2.55 | 2.58 | 2.53 | 2.62 | 2.64 | 2.61 |
| Italy† | 3.30 | 3.34 | 3.26 | 3.30 | 3.34 | 3.22 |
| France† | 0.87 | 1.27 | 0.73 | 1.22 | 1.29 | 1.14 |
| Spain† | 2.56 | 2.69 | 2.55 | 2.64 | 2.66 | 2.62 |
| Belgium† | 1.18 | 1.19 | 1.17 | 1.56 | 1.67 | 1.55 |
| Portugal | 2.49 | 2.51 | 2.47 | 2.24 | 2.26 | 2.23 |
| Finland† | 5.97 | 7.43 | 4.53 | 4.78 | 5.50 | 4.06 |
| Austria† | 2.58 | 4.00 | 1.18 | 1.87 | 2.57 | 1.17 |
| Hungary | 2.65 | 2.68 | 2.56 | 3.26 | 3.33 | 3.09 |
| Cyprus | 3.48 | 3.48 | 3.48 | 3.48 | 3.48 | 3.48 |
| Slovenia | 0.47 | 0.60 | 0.35 | 0.49 | 0.57 | 0.46 |
| Germany† | 2.77 | 3.87 | 1.67 | 2.16 | 2.75 | 1.57 |
| Croatia | −0.36 | −0.36 | −0.36 | −0.21 | −0.19 | −0.21 |
| Slovakia | 0.89 | 0.99 | 0.79 | 1.20 | 1.29 | 1.10 |
| Poland | 0.74 | 1.05 | 0.44 | 1.18 | 1.21 | 1.10 |
| Malta | −0.87 | −0.81 | −1.00 | −0.04 | −0.02 | −0.30 |
| Romania† | 0.54 | 0.72 | 0.36 | 0.79 | 0.88 | 0.70 |
| Netherlands | 0.07 | 0.11 | 0.03 | −0.03 | 0.01 | −0.07 |
| Czechia | 0.38 | 0.42 | 0.34 | 0.66 | 0.70 | 0.62 |
| Latvia | −0.14 | −0.13 | −0.15 | 0.17 | 0.18 | 0.16 |
| Ireland | −0.66 | −0.62 | −0.69 | −0.64 | −0.61 | −0.68 |
| Lithuania | 0.25 | 0.28 | 0.22 | 0.09 | 0.12 | 0.06 |
| Sweden | −0.75 | −0.73 | −0.78 | −0.64 | −0.62 | −0.66 |
| Luxembourg | −0.23 | −0.20 | −0.25 | −0.32 | −0.29 | −0.35 |
| Denmark | −1.09 | −1.09 | −1.09 | −1.78 | −1.75 | −1.82 |
| Bulgaria | −0.89 | −0.88 | −0.89 | −0.53 | −0.52 | −0.53 |
| Estonia | −0.29 | −0.29 | −0.29 | −0.29 | −0.29 | −0.29 |

Source: Author's calculations. Note: Real growth in 2025 shifted by the stated amount with potential growth unchanged, after which the output gap deviation closes via the Commission's two-thirds rule (2024 data and method). Countries are ordered by debt in 2024. †Medium-term plan with a seven-year adjustment period.

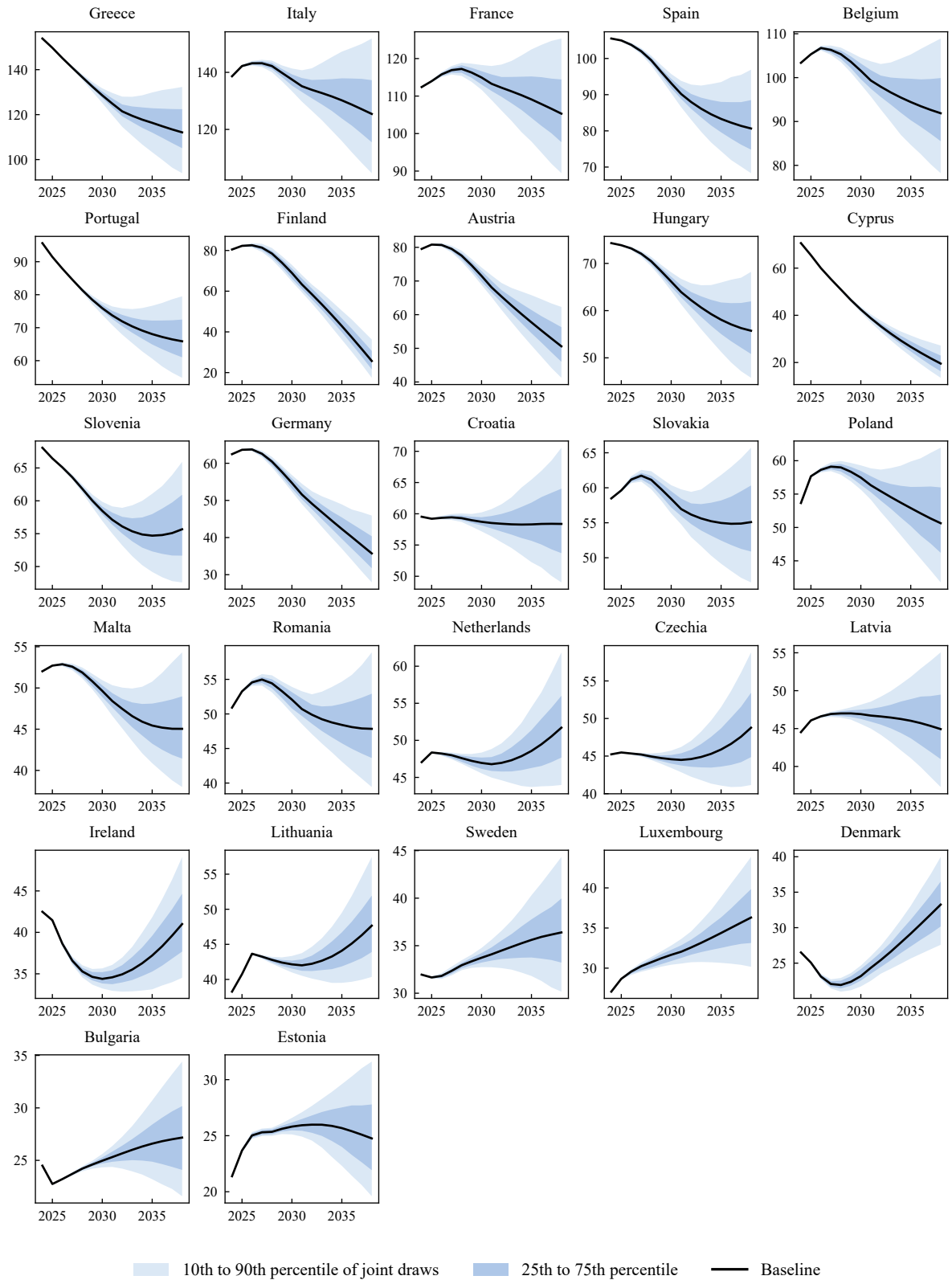



**Figure A1:** Debt under the baseline SPB path in the joint draws, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: Debt when the SPB follows the path of the baseline binding target in all draws, with the adjustment from 2025 to 2028 including the minimum steps of the EDP and the deficit resilience safeguard. The black line shows the baseline; shaded areas show the range between the 10th and 90th percentiles (light) and between the 25th and 75th percentiles (dark) across the 500 joint draws of forecasts and model parameters. Countries are ordered by debt in 2024.

**Table A10:** Criteria that set the targets and largest effects by country, four-year adjustment (percent of GDP and percentage points)

| | Debt | Required | DSA criterion | Binding constraint | Joint draws | | Largest effects on the DSA target |
|---|---|---|---|---|---|---|---|
| | 2024 | adjustment | | | P25 | P75 | |
| Greece | 153.9 | 0.83 | Adverse $r-g$ test: debt declines | Deficit resilience | −0.63 | +1.13 | Threshold 80%, early start, 10-year horizon (+2.05); Threshold 80%, early start, 5-year horizon (+1.55); 2021 Ageing Report (−0.99) |
| Italy | 138.6 | 4.35 | Adverse $r-g$ test: debt declines | DSA criteria | −0.44 | +1.34 | Multiplier 1.5 (+1.36); Potential growth −0.75 pp. (+0.91); Ageing period 0 years (−0.57) |
| France | 112.4 | 3.84 | Deficit below 3% | DSA criteria | −0.07 | +1.07 | Multiplier 1.5 (+2.89); Potential growth −0.75 pp. (+0.67); Multiplier 1 (+0.64) |
| Spain | 105.5 | 3.41 | Adverse $r-g$ test: debt declines | DSA criteria | −0.71 | +0.84 | Ageing period 0 years (−1.83); 2021 Ageing Report (−1.37); Ageing period 5 years (−0.89) |
| Belgium | 103.4 | 3.21 | Adverse $r-g$ test: debt declines | DSA criteria | −0.42 | +0.97 | Multiplier 1.5 (+1.24); 2021 Ageing Report (+0.74); Ageing period 0 years (−0.59) |
| Portugal | 95.6 | 0.28 | Adverse $r-g$ test: debt declines | DSA criteria | −0.67 | +0.67 | Ageing period 0 years (−1.43); Ageing period 5 years (−0.77); 2021 Ageing Report (−0.60) |
| Finland | 80.5 | 6.50 | Stochastic | Debt safeguard | −0.27 | +0.57 | Threshold 80%, early start, 5-year horizon (+0.55); Persistence 5 years (+0.49); AR bootstrap, intercept (+0.46) |
| Austria | 79.5 | 3.92 | Adverse $r-g$ test: debt declines | Debt safeguard | −0.55 | +0.83 | No winsorisation (+0.66); Ageing period 0 years (−0.63); 2021 Ageing Report (+0.48) |
| Hungary | 74.3 | 2.62 | Adverse $r-g$ test: debt declines | DSA criteria | −0.55 | +0.49 | Ageing period 0 years (−0.60); Ageing period 5 years (−0.45); Potential growth −0.75 pp. (+0.36) |
| Slovenia | 68.1 | 1.72 | Lower-SPB test: debt below 60% | DSA criteria | −0.25 | +0.52 | 2021 Ageing Report (+1.01); Ageing period 0 years (−0.79); Potential growth −0.75 pp. (+0.29) |
| Germany | 62.5 | 3.78 | Lower-SPB test: debt below 60% | Debt safeguard | −0.53 | +0.48 | Ageing period 0 years (−0.69); Potential growth +0.75 pp. (−0.39); Potential growth −0.75 pp. (+0.31) |
| Croatia | 59.5 | 1.60 | Baseline: debt below 60% | Deficit resilience | −0.33 | +0.49 | Potential growth −0.75 pp. (+0.29); Potential growth +0.75 pp. (−0.29); Long-term rate −1 pp. (−0.19) |
| Slovakia | 58.5 | 5.15 | Lower-SPB test: debt below 60% | DSA criteria | −0.32 | +0.47 | Ageing period 0 years (−0.65); 2021 Ageing Report (+0.38); Multiplier 1.5 (+0.31) |
| Poland | 53.7 | 3.36 | Deficit below 3% | DSA criteria | −0.14 | +0.34 | Multiplier 1.5 (+1.57); Multiplier 1 (+0.40); Impulse-response rule, 2 years (−0.35) |
| Malta | 52.0 | 2.03 | Deficit below 3% | EDP minimum adjustment | −0.10 | +0.24 | Multiplier 1.5 (+0.73); Impulse-response rule, 2 years (−0.23); Multiplier 1 (+0.19) |
| Romania | 50.9 | 4.95 | Deficit below 3% | DSA criteria | −0.16 | +0.34 | Multiplier 1.5 (+1.15); 2021 Ageing Report (+0.57); Impulse-response rule, 2 years (−0.41) |
| Netherlands | 47.1 | 0.71 | Deficit below 3% | DSA criteria | −0.48 | +0.46 | Ageing period 0 years (−1.54); 2021 Ageing Report (+0.90); Ageing period 5 years (−0.65) |
| Czechia | 45.2 | 0.47 | Deficit below 3% | DSA criteria | −0.46 | +0.42 | Ageing period 0 years (−1.23); Ageing period 5 years (−0.67); Ageing costs: +0.5 pp. (+0.29) |
| Latvia | 44.5 | 1.22 | Deficit below 3% | Deficit resilience | −0.18 | +0.50 | Ageing period 0 years (+0.29); Long-term rate +1 pp. (+0.24); Ageing costs: +0.5 pp. (+0.23) |
| Lithuania | 38.3 | 0.62 | Deficit below 3% | DSA criteria | −0.47 | +0.42 | Ageing period 0 years (−1.39); 2021 Ageing Report (−0.58); Ageing period 5 years (−0.54) |
| Bulgaria | 24.5 | 1.20 | Deficit below 3% | Deficit resilience | −0.25 | +0.35 | 2021 Ageing Report (+0.91); Ageing period 0 years (+0.69); Ageing costs: +0.5 pp. (+0.28) |

Source: Author's calculations. Note: The 21 countries with adjustment needs, ordered by debt in 2024. Required adjustment: baseline binding target minus the SPB in 2024. Joint draws: 25th and 75th percentiles of the change in the DSA target across 500 joint draws of forecasts and model parameters. Largest effects: the three individual checks and alternative settings with the largest absolute change in the DSA target, excluding the lenient and strict composite bundles.